Method of Differentiation

DISCLAIMER

"The content provided herein are created and owned by various authors and licensed to Sorting Hat Technologies Private Limited ("Company"). The Company disclaims all rights and liabilities in relation to the content. The author of the content shall be solely responsible towards, without limitation, any claims, liabilities, damages or suits which may arise with respect to the same."

Method of Differentiation

INTRODUCTION :

The derivative is the instantaneous rate of change of a function with respect to one of its variables. This is equivalent to finding the slope of tangent line to function at a point.

DERIVATIVE BY FIRST PRINCIPLE (DELTA METHOD) :

Let $P(x_0, f(x))$ and $Q(x_0 + h, f(x_0 + h))$ be two points very close to each other on the curve $y = f(x)$. Draw PM and QN perpendicular from P and Q on x-axis, and draw PL as perpendicular from P on QN. Let the chord PQ produced meet their x-axis at R and $\angle QRN = \angle QPL = \phi$.

Now in right-angled triangle QPL

$$\tan\phi = \frac{QL}{PL} = \frac{NQ - NL}{MN} = \frac{NQ - MP}{ON - OM} = \frac{f(x_0 + h) - f(x_0)}{(x_0 + h) - x_0}$$

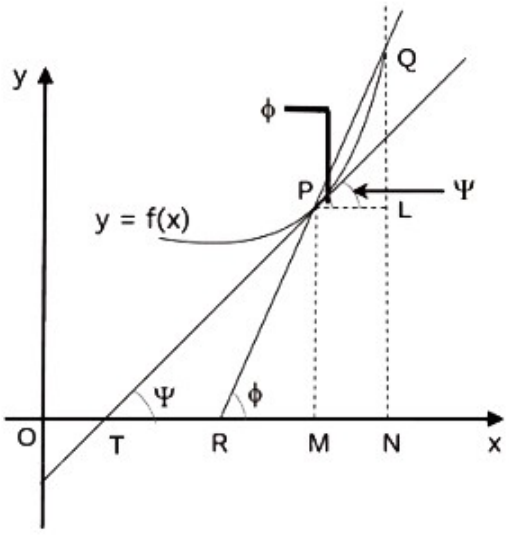
$$= \frac{f(x_0 + h) - f(x_0)}{h} \qquad \ldots (1)$$

where $h \to 0$, the point Q moving along the curve tends to P, i.e., $Q \to P$. The chord PQ approaches the tangent line PT at the point P and then $\phi \to \psi$ Now applying $\lim_{h\to 0}$ in equation (1), we get.

$$\lim_{h\to 0} \tan\phi = \frac{f(x_0 + h) - f(x_0)}{h}$$

$$\tan\psi = \lim_{h\to 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

$$\Rightarrow \quad f'(x_0) = \frac{dy}{dx}\Big|_{x=x_0} = \lim_{h\to 0} \frac{f(x_0 + h) - f(x_0)}{h}$$



$\frac{dy}{dx}$ denotes the instantaneous rate of change of function and is called finding the derivative **by first principle/by delta method/by ab initio/by fundamental definition of calculus**.

Q. **Find equation of tangent to the curve $y = x^2$ at (3, 9).**

Sol. $y = x^2$

$$\frac{dy}{dx} = 2x \qquad \left(\frac{d}{dx}(y^n) = ny^{n-1}\right)$$

$$\Rightarrow \quad \frac{dy}{dx}\Big|_{x=3} = 2(3) = 6$$

Thus, equation of tangent is given by :

$$\Rightarrow \quad y - y_1 = \frac{dy}{dx}\Big|_{x=x_1}(x - x_1)$$

$$\Rightarrow \quad y - 9 = 6(x - 3)$$

$$\Rightarrow \quad \mathbf{y = 6x - 9}$$

Point to Remember!!!

If $y = f(x)$ then the symbols $\frac{dy}{dx} = Dy = f'(x) = y_1$ or y' have the same meaning, which is derivative of y w.r.t. x.

DERIVATIVE OF STANDARD FUNCTIONS :

(1) $\frac{d}{dx}x^n = nx^{n-1}, n \in R$

(2) $\frac{d}{dx}(e^x) = e^x$

(3) $\frac{d}{dx}(a^x) = a^x \ln a, \ a > 0$

(4) $\frac{d}{dx}(\ln x) = \frac{1}{x}$

(5) $\frac{d}{dx}(\sin x) = \cos x$

(6) $\frac{d}{dx}(\cos x) = -\sin x$

(7) $\frac{d}{dx}(\tan x) = \sec^2 x$

(8) $\frac{d}{dx}(\cot x) = -\text{cosec}^2 x$

(9) $\frac{d}{dx}(\sec x) = \sec x \tan x$

(10) $\frac{d}{dx}(\text{cosec}\, x) = -\text{cosec}\, x \cot x$

(11) $\frac{d}{dx}(\sin^{-1}x) = \frac{1}{\sqrt{1 - x^2}}$

(12) $\frac{d}{dx}(\cos^{-1}x) = -\frac{1}{\sqrt{1-x^2}}$

(13) $\frac{d}{dx}(\tan^{-1}x) = \frac{1}{1+x^2}$

(14) $\frac{d}{dx}(\cot^{-1}x) = -\frac{1}{1+x^2}$

(15) $\frac{d}{dx}(\sec^{-1}x) = \frac{1}{|x|\sqrt{x^2-1}}$

(16) $\frac{d}{dx}(\operatorname{cosec}^{-1}x) = -\frac{1}{|x|\sqrt{x^2-1}}$

THEOREM ON DERIVATIVES :

T-1 :

$$\frac{d}{dx}(f_1(x) \pm f_2(x)) = \frac{d}{dx}f_1(x) \pm \frac{d}{dx}f_2(x)$$

T-2 :

$\frac{d}{dx}(kf(x)) = k\frac{d}{dx}f(x)$,
where k is any constant.

T-3 : Chain Rule of derivative

If y = f(u) and u = g(x), where y is a composite function, then $\frac{dy}{dx} = \frac{dy}{du}.\frac{du}{dx}$

T-4 : Product Rule of derivative

If u = f(x) and v = g(x),
then $\frac{d(uv)}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$

T-5 : Quotient of derivative

If u = f(x) and v = g(x),
then $\frac{d\left(\frac{u}{v}\right)}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$

Q.1 If $y=\left(1+x^{\frac{1}{4}}\right)\left(1+x^{\frac{1}{2}}\right)\left(1-x^{\frac{1}{4}}\right)$, then find $\frac{dy}{dx}$.

Sol. $y=\left(1+x^{\frac{1}{4}}\right)\left(1+x^{\frac{1}{2}}\right)\left(1-x^{\frac{1}{4}}\right)=\left(1+x^{\frac{1}{4}}\right)\left(1-x^{\frac{1}{4}}\right)\left(1+x^{\frac{1}{2}}\right)=\left(1-x^{\frac{1}{2}}\right)\left(1+x^{\frac{1}{2}}\right)$

$y=1-x$

$\frac{dy}{dx}=-1$

Q.2 **Find the derivative of the following functions with respect to x :**
(i) $e^{\sqrt{x}}$ **(ii)** xe^x **(iii)** $x^2\ln x$ **(iv)** π^x **(v)** x^π **F**

Sol. **(i)** Let $t=\sqrt{x}$

Using chain rule of derivative,

$$\frac{d}{dx}\left(e^t\right)=\frac{d}{dt}\left(e^t\right).\frac{dt}{dx}$$

$$=e^{\sqrt{x}}\frac{d}{dx}\left(x^{\frac{1}{2}}\right)$$

$$=e^{\sqrt{x}}\frac{1}{2}x^{-\frac{1}{2}}=\frac{\mathbf{e^{\sqrt{x}}}}{\mathbf{2\sqrt{x}}}$$

(ii) Using product rule of derivative,

$$\frac{d}{dx}\left(xe^x\right)=x\frac{d}{dx}\left(e^x\right)+e^x\frac{d}{dx}(x)$$

$$=x\cdot e^x+e^x\cdot 1$$

$$=\mathbf{(x+1)e^x}$$

(iii) Using product rule of derivative,

$$\frac{d}{dx}\left(x^2\ln x\right)=x^2\frac{d}{dx}(\ln x)+\ln x\frac{d}{dx}\left(x^2\right)$$

$$=x^2\cdot\frac{1}{x}+\ln x\cdot 2x$$

$$=x\left(1+2\ln x\right)$$

(iv) $\frac{d}{dx}\left(\pi^x\right)=\pi^x\ln\pi\left(\because D\left(a^x\right)=a^x\ln a\right)$

(v) $\frac{d}{dx}\left(x^\pi\right)=\pi x^{\pi-1}\left(\because Dx^n=nx^{n-1}\right)$

Q.3 **Find the derivative of the following functions with respect to x:**

(i) $\frac{x}{x^2+1}$ **(ii)** $\cos^2 x$ **(iii)** $\sin 3x$ **(iv)** $\sin^{-1}x^2$

(v) $y = x^3 - 3^x$ **(vi)** $3\sin x$ **(vii)** $\ln^2 x$

Sol. **(i)** $y = \frac{x}{x^2+1}$

$$\frac{dy}{dx} = \frac{(x^2+1)\cdot 1 - x(2x+0)}{(x^2+1)^2}$$

$$= \frac{1-x^2}{(x^2+1)^2}$$

(ii) $y = \cos^2 x$

Let $\cos x = t$, then

Using chain rule of derivative,

$$\frac{dy}{dx} = \frac{dy}{dt}\cdot\frac{dt}{dx}$$

$$= \frac{d}{dt}(t^2)\cdot\frac{d}{dx}(\cos x)$$

$$= 2(\cos x)(-\sin x)$$

$$= -2\sin x\cos x$$

$$= -\sin 2x$$

(iii) $y = \sin 3x$

Let $t = 3x$, then

Using chain rule of derivative,

$$\frac{dy}{dx} = \frac{dy}{dt}\cdot\frac{dt}{dx}$$

$$= \frac{d}{dt}(\sin t)\cdot\frac{d}{dt}(3x)$$

$$= 3\cos 3x$$

(iv) Using chain rule of derivatives,

$$\frac{dy}{dx} = \frac{d}{dx}\left(\sin^{-1}x^2\right)\cdot\frac{d}{dx}\left(x^2\right)$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-(x^2)^2}}\times 2x$$

$$\Rightarrow \frac{dy}{dx} = \frac{2x}{\sqrt{1-x^4}}$$

(v) $y = x^3 - 3^x$

$$\frac{dy}{dx} = \frac{d}{dx}(x^3) - \frac{d}{dx}(3^x)$$

$$\frac{dy}{dx} = 3x^2 - 3^x \ln 3$$

(vi) Using product rule of derivative,

$$\frac{dy}{dx} = 3\frac{d}{dx}(\sin x) + \sin x \frac{d}{dx}(3)$$

$$\frac{dy}{dx} = 3\cos x + \sin x(0) = 3\cos x$$

(vii) $y = \ln^2 x$

Let $\ln x = \theta$ then, using chain rule,

$$\frac{dy}{dx} = \frac{d}{d\theta}(\theta^2) \cdot \frac{d\theta}{dx}$$

$$= 2(\ln x) \cdot \frac{d}{dx}(\ln x)$$

$$= 2\ln x \cdot \left(\frac{1}{x}\right)$$

$$= \frac{2\ln x}{x}$$

Q.4 **Find derivatives of following functions w.r.t. x**

(i) $\tan(\tan^{-1} x)$ **(ii)** $\cos^4 \frac{x}{2} - \sin^4 \frac{x}{2}$ **(iii)** $(\cos^{-1} x + \sin^{-1} x)^n$

(iv) $e^{\ln\cot^{-1}x}$ **(v)** $\frac{1 - \cos 2x}{\sin 2x}$ **(vi)** $\tan^{-1} x + \tan^{-1}\left(\frac{1}{x}\right)$

(vii) $x\sin^{-1}x$ **(viii)** $e^x . \tan^{-1} x$

Sol. **(i)** $\tan(\tan^{-1} x) = x$

$$\frac{d}{dx}(\tan(\tan^{-1} x)) = \frac{d}{dx}(x) = 1$$

(ii) $\cos^4 \frac{x}{2} - \sin^4 \frac{x}{2} = \left(\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}\right)\left(\cos^2 \frac{x}{2} + \sin^2 \frac{x}{2}\right)$

$$= \cos^2 \frac{x}{2} - \sin^2 \frac{x}{2} = \cos x$$

$$\frac{d}{dx}\left(\cos^4 \frac{x}{2} - \sin^4 \frac{x}{2}\right) = \frac{d}{dx}\left(\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}\right)$$

$$= \frac{d}{dx}(\cos x) = -\sin x$$

(iii) $\cos^{-1} x + \sin^{-1} x = \frac{\pi}{2}$

$$\frac{d}{dx}\left(\cos^{-1} x + \sin^{-1} x\right)^n = \frac{d}{dx}\left(\frac{\pi}{2}\right)^n = 0$$

(iv) $e^{\ln \cot^{-1} x} = \cot^{-1} x \qquad \left(\text{Using } a^{\log_a N} = N\right)$

$$\frac{dy}{dx} = \frac{d}{dx}\left(\cot^{-1} x\right)$$

$$\frac{dy}{dx} = -\frac{1}{1+x^2}$$

(v) $$\frac{d}{dx}\left(\frac{1-\cos 2x}{\sin 2x}\right) = \frac{d}{dx}\left(\frac{2\sin^2 x}{2\sin x \cos x}\right) = \frac{d}{dx}(\tan x) = \sec^2 x$$

(vi) $$\tan^{-1} x + \tan^{-1}\left(\frac{1}{x}\right) = \begin{cases} \frac{\pi}{2}, & x > 0 \\ -\frac{\pi}{2}, & x < 0 \end{cases}$$

$$\frac{dy}{dx} = \begin{cases} 0, & x > 0 \\ 0, & x < 0 \end{cases}$$

(vii) Using product rule:

$$\frac{dy}{dx} = \frac{d}{dx}\left(x \sin^{-1} x\right)$$

$$\frac{dy}{dx} = \sin^{-1} x\,(1) + x\left(\frac{1}{\sqrt{1-x^2}}\right)$$

$$\frac{dy}{dx} = \sin^{-1} x + \frac{x}{\sqrt{1-x}}$$

(viii) Using product rule,

$$\frac{dy}{dx} = \frac{d}{dx}\left(e^x \cdot \tan^{-1} x\right)$$

$$\frac{dy}{dx} = e^x \cdot \tan^{-1} x + e^x \cdot \left(\frac{1}{1+x^2}\right)$$

$$\frac{dy}{dx} = e^x\left[\tan^{-1} x + \frac{1}{1+x^2}\right]$$

Note :

If 3 functions are involved then remember

$$D(f(x).g(x).h(x)) = f(x).g(x).h'(x) + g(x).h(x).f'(x) + h(x).f(x).g'(x)$$

$$= \frac{(fg)'(h) + (gh)'(f) + (hf)'(g)}{2}$$

This result can be generalised to the product of n functions,

Proof :

$$\frac{d}{dx}[(fg)h] = (fg)'h + fgh' = (f'g + fg')h + fgh' = f'gh + fg'h + fgh'$$

Similarly, $\frac{d}{dx}(fghk) = f'ghk + fg'hk + fgh'k + fghk'$

Now, $\frac{(fg)'h + (gh)'f + (hf)'g}{2}$

$$= \frac{(f'g + g'f)h + (g'h + gh')f + (h'f + hf')g}{2}$$

$$= \frac{2(f'gh + fg'h + fgh')}{2}$$

$$= f'gh + fg'h + fgh'$$

$$= D(fgh)$$

Q.1 **Let $F(x) = f(x).g(x).h(x)$. If some $x = x_0$, $F'(x_0) = 2021F(x_0)$; $f'(x_0) = 4f(x_0)$, $g'(x_0) = -7g(x_0)$ and $h'(x_0) = kh(x_0)$. Then find k.**

Sol. $F(x) = f(x)\, g(x)\, h(x)$

Given that $F'(x_0) = 2021F(x_0), f'(x_0) = 4f(x_0); g'(x_0) = -7g(x_0)$ and

$h'(x_0) = kh(x_0)$

$F'(x_0) = f'(x_0)g(x_0)h(x_0) + f(x_0)g'(x_0)h(x_0) + f(x_0)g(x_0)h'(x_0)$

$2021F(x_0) = (4 - 7 + k)f(x_0)g(x_0)h(x_0)$

$2021 = -3 + k \Rightarrow k = 2024$

Q.2 If $f(x) = (1+x)(3+x^2)^{1/2}(9+x^3)^{1/3}$ then $f'(-1)$ is equal to

(A) 0 **(B)** $2\sqrt{2}$ **(C) 4** **(D) 6**

Sol. **(C)**

Let $(1+x) = g(x)$ and $(3+x^2)^{\frac{1}{2}}(9+x^3)^{\frac{1}{3}} = k(x)$

$\Rightarrow f(x) = g(x).k(x)$

$\Rightarrow f'(x) = g'(x).k(x) + g(x)k'(x)$

At $x = -1$, $g(-1) = 0$

$\Rightarrow f'(-1) = g'(-1)k(-1) + 0 = 1 \cdot k(-1)$ $\quad (\because g'(x) = 1)$

$\Rightarrow f'(-1) = k(-1) = \left(3+(-1)^2\right)^{\frac{1}{2}} \cdot \left(9+(-1)^3\right)^{\frac{1}{3}}$

$\Rightarrow f'(-1) = 4^{\frac{1}{2}} \cdot 8^{\frac{1}{3}} = 2 \times 2 = 4$

Q.3 **Let f, g and h are differentiable functions. If f(0) = 1; g(0) = 2; h(0) = 3 and the derivatives of their pair wise products at x=0 are** $(fg)'(0) = 6; (gh)'(0) = 4$ **and** $(hf)'(0) = 5$, **then compute the value of** $(fgh)'(0)$.

Sol. $(fgh)'(0) = \dfrac{(fg)'(0).h(0) + (gh)'(0).f(0) + (hf)'(0).g(0)}{2}$

$\Rightarrow (fgh)'(0) = \dfrac{6\times 3 + 4\times 1 + 5\times 2}{2}$

$\Rightarrow (fgh)'(0) = 16$

Q.4 **If** $f(x) = 1 + x + x^2 + \ldots + x^{100}$ **then** $f'(1) = ?$

Sol. $f(x) = 1 + x + x^2 + \ldots + x^{100}$

$\Rightarrow f'(x) = 0 + 1 + 2x + \ldots + 100x^{99}$

$\Rightarrow f'(1) = 1 + 2 + \ldots + 100 = \dfrac{100(100+1)}{2} = 5050$

Q.5 **If** $f(x) = 1 + x + x^2 + \ldots + x^{100}$ **then** $f''''(0) = ?$

Sol. $f'(x) = 0 + 1 + 2x + 3x^2 + 4x^3 + \ldots + 100x^{99}$

$\Rightarrow f''(x) = 0 + 0 + 2 + 6x + 12x^2 + \ldots + 100 \times 99.x^{98}$

$\Rightarrow \quad f'''(x) = 0 + 0 + 0 + 6 + 24x + ... + 100 \times 99 \times 98.x^{97}$

$\Rightarrow \quad f''''(x) = 0 + 0 + 0 + 0 + 24 + ... + 100 \times 99 \times 98 \times 97.x^{96}$

$\Rightarrow \quad f''''(0) = 24$

Q.6 **If $y = \frac{1 - \ln x}{1 + \ln x}$ then value of $\frac{dy}{dx}$ at $x = e$ is ___.**

Sol. Using quotient rule :

$$\frac{dy}{dx} = \frac{(1+\ln x)\left(0 - \frac{1}{x}\right) - (1 - \ln x)\left(0 + \frac{1}{x}\right)}{(1+\ln x)^2} = \frac{-\frac{1}{x}[1 + \ln x + 1 - \ln x]}{(1+\ln x)^2}$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{-2}{x(1+\ln x)^2}$$

Thus, $\frac{dy}{dx}\Big|_{x=e} = \frac{-2}{e(1+1)^2} = \frac{-1}{2e}$

Q.7 **If $y = \frac{\sin^{-1}x}{\cos^{-1}x}$ then value of $\frac{dy}{dx}$ at $x = \frac{1}{\sqrt{2}}$ is ___.**

Sol. $$y = \frac{\sin^{-1}x}{\cos^{-1}x} = \frac{\frac{\pi}{2} - \cos^{-1}x}{\cos^{-1}x} = \frac{\frac{\pi}{2}}{\cos^{-1}x} - 1$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{\pi}{2}\left[\frac{-1}{(\cos^{-1}x)^2}\right] \times \frac{-1}{\sqrt{1-x^2}}$$

$$\Rightarrow \quad \frac{dy}{dx}\Big|_{x=\frac{1}{\sqrt{2}}} = \frac{\pi}{2} \times \frac{-1}{\left(\cos^{-1}\frac{1}{\sqrt{2}}\right)^2} \times \frac{-1}{\sqrt{1-\left(\frac{1}{\sqrt{2}}\right)^2}} = \frac{\pi}{2} \times \frac{1}{\left(\frac{\pi}{4}\right)^2} \times \sqrt{2} = \frac{\pi}{2} \times \frac{16}{\pi^2} \times \sqrt{2}$$

$$\Rightarrow \quad \frac{dy}{dx}\Big|_{x=\frac{1}{\sqrt{2}}} = \frac{8\sqrt{2}}{\pi}$$

Q.8 **If $y = \frac{x^3 + 2^x}{e^x}$ then $\frac{dy}{dx} = ?$**

Sol. $$y = \frac{x^3 + 2^x}{e^x} = x^3.e^{-x} + \left(\frac{2}{e}\right)^x$$

$$\frac{dy}{dx} = 3x^2.e^{-x} - e^{-x}.x^3 + \left(\frac{2}{e}\right)^x \ln\left(\frac{2}{e}\right)$$

Q.9 If $y = \frac{x\sin x}{1+\tan x}$ then $\frac{dy}{dx} = ?$

Sol. Using quotient rule:

$$\frac{dy}{dx} = \frac{(1+\tan x)(\sin x \cdot 1 + x\cos x) - (x\sin x)\sec^2 x}{(1+\tan x)^2}$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{(1+\tan x)(\sin x + x\cos x) - x\sin x\sec^2 x}{(1+\tan x)^2}$$

Q.10 If $y = \frac{x^4+x^2+1}{x^2+x+1}$ then $\frac{dy}{dx} = ax + b$. **Find a and b.**

Sol. $$y = \frac{x^4+x^2+1}{x^2+x+1} = \frac{(x^2+x+1)(x^2-x+1)}{x^2+x+1}$$

$$\Rightarrow \quad y = x^2 - x + 1$$

$$\Rightarrow \quad \frac{dy}{dx} = 2x - 1 = ax + b$$

on comparing, **a = 2** and **b = -1**

Q.11 If $y = \frac{\sec x + \tan x - 1}{\tan x - \sec x + 1}$, **find** $\frac{dy}{dx}\Big|_{x=\frac{\pi}{4}}$.

Sol. $$y = \frac{\sec x + \tan x - 1}{\tan x - \sec x + 1} = \frac{\sec x + \tan x - (\sec^2 x - \tan^2 x)}{\tan x - \sec x + 1}$$

$$\Rightarrow \quad y = \frac{(\sec x + \tan x) - (\sec x - \tan x)(\sec x + \tan x)}{\tan x - \sec x + 1}$$

$$\Rightarrow \quad y = \frac{(\sec x + \tan x)[1 - (\sec x - \tan x)]}{\tan x - \sec x + 1}$$

$$\Rightarrow \quad y = \sec x + \tan x$$

$$\Rightarrow \quad \frac{dy}{dx} = \sec x \tan x + \sec^2 x$$

$$\Rightarrow \quad \frac{dy}{dx} = \sec x(\tan x + \sec x)$$

Thus, $\frac{dy}{dx}\Big|_{x=\frac{\pi}{4}} = \sqrt{2}(1+\sqrt{2}) = 2 + \sqrt{2}$

Q.12 If $y = \dfrac{\tan^{-1}x - \cot^{-1}x}{\tan^{-1}x + \cot^{-1}x}$, find $\left.\dfrac{dy}{dx}\right]_{x=-1}$

(A) 0 **(B) 1** **(C) $\dfrac{2}{\pi}$** **(D) –1**

Sol. **(C)**

$$y = \frac{\tan^{-1}x - \cot^{-1}x}{\tan^{-1}x + \cot^{-1}x} = \frac{\tan^{-1}x - \left(\frac{\pi}{2} - \tan^{-1}x\right)}{\frac{\pi}{2}} = \frac{2}{\pi}\left(2\tan^{-1}x\right) - 1$$

$$y = \frac{4}{\pi}\tan^{-1}x - 1$$

$$\frac{dy}{dx} = \frac{4}{\pi}\left(\frac{1}{1+x^2}\right) \Rightarrow \left.\frac{dy}{dx}\right|_{x=-1} = \frac{4}{\pi}\left(\frac{1}{1+(-1)^2}\right) = \frac{4}{\pi} \times \frac{1}{2} = \frac{2}{\pi}$$

Q.13 If $y = \dfrac{x^3 + x^2 + x}{1 + x^2}$, find $\left.\dfrac{dy}{dx}\right]_{x=0}$

Sol.

$$y = \frac{x^3 + x^2 + x}{1 + x^2} = \frac{x^3 + x}{1 + x^2} + \frac{x^2 + 1 - 1}{1 + x^2} = x + 1 - \frac{1}{1 + x^2}$$

$$\Rightarrow \frac{dy}{dx} = 1 + 0 - \left[\frac{-1}{(1+x^2)^2}(2x)\right] = 1 + \frac{2x}{(1+x^2)^2}$$

Thus, $\left.\dfrac{dy}{dx}\right|_{x=0} = 1 + 0 = 1$

Q.14 **Let g be a differentiable function of x. If $f(x) = \dfrac{g(x)}{x^2}$ for x > 0, g(2) = 3 and $g'(2) = -2$, find $f'(2)$.**

Sol.

$$f(x) = \frac{g(x)}{x^2}$$

$$\Rightarrow \quad f'(x) = \frac{x^2 g'(x) - g(x) \cdot 2x}{x^4} \qquad \text{(Using quotient rule)}$$

$$\text{Thus, } f'(2) = \frac{(2)^2 \cdot g'(2) - g(2) \cdot 2(2)}{2^4}$$

$$\Rightarrow \quad f'(2) = \frac{4(-2) - 3(4)}{16} = \frac{4}{16}[-2 - 3]$$

$$\Rightarrow \quad f'(2) = \frac{-5}{4}$$

Note :

If f'(x) is not defined on x = c then it is wrong to conclude that f(x) is not derivable at x = c. In such cases, check LHD at x = c and RHD at x = c.

Q.15 $f(x) = x^{\frac{1}{3}}\sin x$ **then** $f'(0) = ?$

Sol. $f(x) = x^{\frac{1}{3}}\cos x + \frac{1}{3} \times \frac{1}{x^{\frac{2}{3}}}\sin x$

From here, $f'(0)$ is not defined. But this method is incorrect as product rule is only applicable when both the functions are differentiable.

Here, $x^{\frac{1}{3}}$ is not differentiable at x = 0.

From first principle of derivative:

$$f'(0) = \lim_{h\to 0}\frac{f(0+h) - f(0)}{h}$$

$$\Rightarrow \quad f'(0) = \lim_{h\to 0}\frac{h^{\frac{1}{3}}\sinh - 0}{h}$$

$$\Rightarrow \quad f'(0) = \lim_{h\to 0} h^{\frac{1}{3}}\frac{\sinh}{h} = 0$$

Q.16 **If** $y = \sin^3\sqrt{x}$ **then** $\frac{dy}{dx} = ?$

Sol. Let $\sin\sqrt{x} = \theta$

$$\Rightarrow \quad y = \theta^3$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{dy}{d\theta}.\frac{d\theta}{dx} = 3\sin^2\left(\sqrt{x}\right).\frac{d}{dx}\left(\sin\sqrt{x}\right)$$

$$\Rightarrow \quad \frac{dy}{dx} = 3\sin^2\left(\sqrt{x}\right)\cos\sqrt{x}.\frac{d}{dx}\left(\sqrt{x}\right)$$

$$\Rightarrow \quad \frac{dy}{dx} = 3\sin^2\left(\sqrt{x}\right)\cos\sqrt{x}.\frac{1}{2\sqrt{x}}$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{3}{2}.\frac{\sin^2\sqrt{x}\cos\sqrt{x}}{\sqrt{x}}$$

Q.17 If $y = \ln^3\left(\tan^2\left(x^4\right)\right)$ then $\frac{dy}{dx} = ?$

Sol. Let $\tan^2\left(x^4\right) = \theta$

$$\Rightarrow \quad y = \ln^3\theta$$

$$\Rightarrow \quad \frac{dy}{dx} = 3\ln^2\left(\tan^2\left(x^4\right)\right) \cdot \frac{d}{dx}\left[\ln\left(\tan^2\left(x^4\right)\right)\right]$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{3\ln^2\left(\tan^2\left(x^4\right)\right)}{\tan^2\left(x^4\right)} \cdot \frac{d}{dx}\left(\tan^2\left(x^4\right)\right)$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{3\ln^2\left(\tan^2\left(x^4\right)\right)}{\tan^2\left(x^4\right)} \cdot 2\tan\left(x^4\right) \cdot \frac{d}{dx}\left(\tan\left(x^4\right)\right)$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{24\ln^2\left(\tan^2\left(x^4\right)\right)}{\tan\left(x^4\right)} \cdot \sec^2\left(x^4\right) \cdot x^3$$

Q.18 If $y = \cos^{-1}\left(\frac{ax}{b}\right)$ then $\frac{dy}{dx} = ?$ **(b > 0)**

Sol. $y = \cos^{-1}\left(\frac{ax}{b}\right)$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{-1}{\sqrt{1-\left(\frac{ax}{b}\right)^2}} \times \frac{d}{dx}\left(\frac{ax}{b}\right)$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{-b}{\sqrt{b^2 - a^2x^2}} \times \frac{a}{b}$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{-a}{\sqrt{b^2 - a^2x^2}}$$

Q.19 If $y = \frac{1}{(f(x))^n}$ then $\frac{dy}{dx} = ?$

Sol. $y = \frac{1}{(f(x))^n} = (f(x))^{-n}$

$$\Rightarrow \quad \frac{dy}{dx} = (-n)(f(x))^{-n-1} \cdot \frac{d}{dx}(f(x))$$

$$\Rightarrow \quad \frac{dy}{dx} = -n\left(f(x)\right)^{-n-1} \cdot f'(x)$$

Q.20 If $y = \ln(\sec x)$, then $\frac{dy}{dx} = ?$

Sol. $y = \ln(\sec x)$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\sec x} \cdot \frac{d}{dx}(\sec x)$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\sec x} \cdot (\sec x \tan x)$$

$$\Rightarrow \frac{dy}{dx} = \tan x$$

Q.21 $y = \sec x\left(\sqrt{\tan x}\right)$

Sol. $y = \sec x \cdot \sqrt{\tan x}$

$$\Rightarrow \frac{dy}{dx} = (\sec x \tan x) \cdot \sqrt{\tan x} + \sec x \cdot \frac{1}{2\sqrt{\tan x}} \cdot \left(\sec^2 x\right)$$

$$\Rightarrow \frac{dy}{dx} = \sec x\left((\tan x)^{\frac{3}{2}} + \frac{\sec^2 x}{2\sqrt{\tan x}}\right)$$

Q.22 If $y = \sec^2\left(f^3(x)\right)$, then $\frac{dy}{dx} = ?$

Sol. $y = \sec^2\left(f^3(x)\right)$

$$\Rightarrow \frac{dy}{dx} = 2\sec\left(f^3(x)\right) \times \sec\left(f^3(x)\right).\tan\left(f^3(x)\right) \times \frac{d}{dx}\left(f^3(x)\right)$$

$$\Rightarrow \frac{dy}{dx} = 2\sec^2\left(f^3(x)\right)\tan\left(f^3(x)\right) \times 3f^2(x).f'(x)$$

Q.23 $y = \sqrt{f(x)}$

Sol. $y = \sqrt{f(x)}$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{2\sqrt{f(x)}} \cdot \frac{d}{dx}(f(x))$$

$$\Rightarrow \frac{dy}{dx} = \frac{f'(x)}{2\sqrt{f(x)}}$$

Q.24 If $y = e^{\cos^2(\tan^{-1}x^3)^2}$, then $\frac{dy}{dx} = ?$

Sol. $y = e^{\cos^2(\tan^{-1}x^3)^2}$

$$\Rightarrow \frac{dy}{dx} = e^{\cos^2(\tan^{-1}x^3)^2} \cdot 3\cos^2\left(\tan^{-1}x^3\right)^2 \cdot \left(-\sin\left(\tan^{-1}x^3\right)^2\right) \cdot 2\tan^{-1}x^3 \cdot \frac{1}{1+\left(x^3\right)^2}.3x^2$$

$$\Rightarrow \frac{dy}{dx} = -18e^{\left(\cos^2(\tan^{-1}x^3)^2\right)} \cdot \cos^2\left(\tan^{-1}x^3\right)^2 \cdot \sin\left(\tan^{-1}x^3\right)^2 \cdot \tan^{-1}x^3 \cdot \frac{x^2}{1+x^6}$$

Q.25 If $y = \cos(\ln x)$, then $\frac{dy}{dx} = ?$

Sol. $y = \cos(\ln x)$

$$\Rightarrow \frac{dy}{dx} = -\sin(\ln x) \cdot \frac{d}{dx}(\ln x) = -\frac{\sin(\ln x)}{x}$$

Q.26 If $y = f\left(\frac{1}{x}\right)$, then $\frac{dy}{dx} = ?$

Sol. $y = f\left(\frac{1}{x}\right)$

$$\Rightarrow \frac{dy}{dx} = f'\left(\frac{1}{x}\right)\frac{d}{dx}\left(\frac{1}{x}\right)$$

$$\Rightarrow \frac{dy}{dx} = f'\left(\frac{1}{x}\right)\left(\frac{-1}{x^2}\right) = \frac{-f'\left(\frac{1}{x}\right)}{x^2}$$

Q.27 **Suppose that f is a differentiable function such that f(2) = 1 and f'(2) = 3 and let g(x) = f(xf(x)). Find g'(2).**

Sol. $g(x) = f(xf(x))$

$$\Rightarrow g'(x) = f'(xf(x)).\left[1.f(x) + xf'(x)\right]$$

$$\Rightarrow g'(2) = f'(2f(2)).\left[f(2) + 2f'(2)\right]$$

$$\Rightarrow g'(2) = f'(2\times 1).[1 + 2\times 3]$$

$$\Rightarrow g'(2) = 3\times 7 = 21$$

Logarithmic Differentiation

(i) A function which is the product or quotient of a number of functions.

$H(x) = \dfrac{f(x)g(x)}{k(x)\ell(x)}$ then we will take logarithm before taking derivative.

$\Rightarrow \ln(H(x)) = \ln(f(x)) + \ln(g(x)) - \ln(k(x)) - \ln(\ell(x))$

And then take derivative of functions on both sides,

$$\Rightarrow \frac{H'(x)}{H(x)} = \frac{f'(x)}{f(x)} + \frac{g'(x)}{g(x)} - \frac{k'(x)}{k(x)} - \frac{\ell'(x)}{\ell(x)}$$

OR

(ii) A function of the form $[f(x)]^{g(x)}$ where f and g are both derivable. It will be convenient to take the logarithm of the function first and then differentiate OR express.

$(f(x))^{g(x)} = e^{g(x)\cdot\ln(f(x))}$ and then differentiate.

Now, $[f(x)]^{g(x)} = e^{\ln\left([f(x)]^{g(x)}\right)} = e^{g(x)\cdot\ln(f(x))}$

Taking derivative on both sides,

$$\Rightarrow \frac{d}{dx}\left\{[f(x)]^{g(x)}\right\} = e^{g(x)\cdot\ln(f(x))}\cdot\frac{d}{dx}\{g(x)\cdot\ln(f(x))\}$$

$$\Rightarrow \frac{d}{dx}\left\{[f(x)]^{g(x)}\right\} = e^{g(x)\cdot\ln(f(x))}\cdot\left\{g'(x)\cdot\ln(f(x)) + g(x)\cdot\frac{1}{f(x)}\cdot\frac{d}{dx}(f(x))\right\}$$

Q.1 **If $y = \sin x\cdot\sin 2x\cdot\sin 3x\cdot\ \ldots\ .\cdot\sin nx$, $n \in N$, $n > 1$. Find y' and $y'(0)$.**

Sol. $y = \sin x\cdot\sin 2x\cdot\sin 3x\cdot.\ \ldots\ \cdot\sin nx$

Taking logarithm on both sides, we get

$\ln y = \ln(\sin x) + \ln(\sin 2x) + \ln(\sin 3x) + \ldots + \ln(\sin nx)$

Now, differentiating on both sides, we get

$$\frac{1}{y}\cdot\frac{dy}{dx} = \frac{1}{\sin x}\cdot\cos x + \frac{1}{\sin 2x}\cdot\cos 2x\cdot 2 + \frac{1}{\sin 3x}\cdot\cos 3x\cdot 3 + \ldots + \frac{1}{\sin nx}\cdot\cos nx\cdot n$$

$$\Rightarrow \frac{dy}{dx} = \{\sin x\cdot\sin 2x\ldots\ \sin nx\}\{\cot x + 2\cot 2x + \ldots\ +\ n\cot nx\}$$

$$\Rightarrow \frac{dy}{dx} = \left\{\prod_{r=1}^{n}\sin(rx)\right\}\left\{\sum_{r=1}^{n} r\cot(rx)\right\}$$

$$\Rightarrow y'(0) = 0$$

Q.2 If $f(x) = (x+1)(x+2)(x+3)\ldots(x+n)$ then $f'(0)$ is

(A) n!

(B) $\left(1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\ldots+\frac{1}{n}\right)$

(C) $\frac{n(n+1)}{2}$

(D) $n!\left(1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\ldots+\frac{1}{n}\right)$

Sol. **(D)**

$y = f(x) = (x+1)(x+2)(x+3)\ldots(x+n)$

$\Rightarrow \quad \ln y = \ln(x+1)+\ln(x+2)+\ln(x+3)+\ldots+\ln(x+n)$

$\Rightarrow \quad \frac{1}{y}\frac{dy}{dx} = \frac{1}{x+1}+\frac{1}{x+2}+\frac{1}{x+3}+\ldots+\frac{1}{x+n}$

$\Rightarrow \quad y' = \left[(x+1)(x+2)\ldots(x+n)\right]\left[\frac{1}{x+1}+\frac{1}{x+2}+\ldots+\frac{1}{x+n}\right]$

$\Rightarrow \quad f'(0) = (1\cdot 2\cdot 3\cdot \ldots\cdot n)\left[1+\frac{1}{2}+\ldots+\frac{1}{n}\right] = n!\left[1+\frac{1}{2}+\frac{1}{3}+\ldots+\frac{1}{n}\right]$

Q.3 If $f(x) = \prod_{n=1}^{100}(x-n)^{r(101-r)}$ then find $\frac{f'(101)}{f(101)}$.

Sol. $f(x) = \prod_{r=1}^{100}(x-n)^{n(101-n)}$

$\Rightarrow \quad f(x) = (x-1)^{100}(x-2)^{2\times 99}\ldots(x-n)^{n(101-n)}$

$\Rightarrow \quad \ln(f(x)) = \ln(x-1)^{100}+\ln(x-2)^{2\times 99}+\ldots+\ln(x-n)^{r(101-n)}$

$\Rightarrow \quad \ln(f(x)) = \sum_{n=1}^{100}\left(\ln(x-n)^{r(101-n)}\right)$

$\Rightarrow \quad \ln(f(x)) = \sum_{n=1}^{100} n(101-n)\log(x-n)$

Taking derivatives on both sides, we get

$\Rightarrow \quad \frac{1}{f(x)}\cdot f'(x) = \sum_{n=1}^{100} n(101-n)\cdot\frac{1}{x-n}$

Now, put x = 101, we get

$\Rightarrow \quad \frac{f'(101)}{f(101)} = \sum_{n=1}^{100} n(101-n)\cdot\frac{1}{101-n}$

$\Rightarrow \quad \frac{f'(101)}{f(101)} = \sum_{n=1}^{100} n = \frac{100\times 101}{2} = 5050$

Q.4 **Find derivative of** $y = (\sin x)^{\ln x}$

Sol. Given $y = (\sin x)^{\ln x}$

Then, $y = e^{\ln x \cdot \ln \sin x}$

Differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = e^{\ln x \cdot \ln \sin x} \frac{d}{dx}\{\ln x \cdot \ln \sin x\}$$

$$\Rightarrow \quad \frac{dy}{dx} = (\sin x)^{\ln x} \left\{\ln \sin x \frac{d}{dx}(\ln x) + \ln x \frac{d}{dx}(\ln \sin x)\right\}$$

$$\Rightarrow \quad \frac{dy}{dx} = (\sin x)^{\ln x} \left\{\frac{\ln \sin x}{x} + \ln x \frac{1}{\sin x} \cos x\right\}$$

$$\Rightarrow \quad \frac{dy}{dx} = (\sin x)^{\ln x} \left\{\frac{\ln \sin x}{x} + \cot x \ln x\right\}$$

Q.5 $y = x^{\tan x} + (\sin x)^{\cos x}$

Sol. $y = x^{\tan x} + (\sin x)^{\cos x}$

Let $u = x^{\tan x}, v = (\sin x)^{\cos x}$

$$\ln u = \tan x \cdot \ln x$$

$$\Rightarrow \quad \frac{d}{dx}(\ln u) = \frac{d}{dx}(\tan x \cdot \ln x)$$

$$\Rightarrow \quad \frac{1}{u}\frac{du}{dx} = \tan x\left(\frac{1}{x}\right) + \sec^2 x \cdot \ln x$$

$$\frac{du}{dx} = x^{\tan x}\left(\frac{\tan x}{x} + \sec^2 x \cdot \ln x\right) \qquad \ldots(i)$$

$$v = (\sin x)^{\cos x}$$

$$\ln v = \cos x \cdot \ln \sin x$$

$$\Rightarrow \quad \frac{d}{dx}\ln v = \frac{d}{dx}(\cos x \cdot \ln \sin x)$$

$$\Rightarrow \quad \frac{1}{v}\frac{dv}{dx} = \left(\cos x \cdot \frac{\cos x}{\sin x}\right) + \ln \sin x \cdot (-\sin x)$$

$$\Rightarrow \quad \frac{dv}{dx} = (\sin x)^{\cos x}\left(\frac{\cos^2 x}{\sin x} - \sin x \cdot \ln(\sin x)\right) \qquad \ldots(ii)$$

$$y = u + v$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$$

From (i) and (ii)

$$\frac{dy}{dx} = x^{\tan x}\left(\frac{\tan x}{x} + \sec^2 x \cdot \ln x\right) + (\sin x)^{\cos x}\left(\frac{\cos^2 x}{\sin x} - \sin x \cdot \ln(\sin x)\right)$$

Q.6 If $y=(\sin x)\left(e^{\sqrt{\sin x}}\right)(\ln x)\left(x^{\cos^{-1}x}\right)$ then find $\frac{dy}{dx}$?

Sol. Taking log on both sides, we get

$$\ln y=\ln\sin x+\sqrt{\sin x}+\ln(\ln x)+\cos^{-1}x\cdot\ln x$$

Differentiating on both sides,

$$\frac{1}{y}\left(\frac{dy}{dx}\right)=\frac{\cos x}{\sin x}+\frac{1}{2\sqrt{\sin x}}\cdot\cos x+\frac{1}{\ln x}\times\frac{1}{x}+\ln x\left(\frac{-1}{\sqrt{1-x^2}}\right)+\left(\cos^{-1}x\cdot\frac{1}{x}\right)$$

$$\frac{dy}{dx}=y\left(\cot x+\frac{\cos x}{2\sqrt{\sin x}}+\frac{1}{x\ln x}-\frac{\ln x}{\sqrt{1-x^2}}+\frac{\cos^{-1}x}{x}\right)$$

Q.7 If $y=\left(x^{\ln x}\right)(\sec x)^{3x}$ then find $\frac{dy}{dx}$?

Sol. $y=\left(x^{\ln x}\right)(\sec x)^{3x}$

$$\ln y=(\ln x)^2+3x\ln(\sec x)$$

$$\frac{1}{y}\frac{dy}{dx}=\frac{2\ln x}{x}+3\ln(\sec x)+\frac{3x}{(\sec x)}(\sec x\tan x)$$

$$\frac{dy}{dx}=y\left(\frac{2\ln x}{x}+3\ln(\sec x)+3x\tan x\right).$$

Q.8 If $y=x^{e^{x^{e^{x}}}}$ then $\frac{dy}{dx}=?$

Sol. $y=e^{\left\{e^{x^{e^{x}}}(\ln x)\right\}}$

Differentiating on both sides, we get

$$\frac{dy}{dx}=y\cdot\frac{d}{dx}\left(e^{x^{e^{x}}}(\ln x)\right)=y\left\{\frac{1}{x}\cdot e^{x^{e^{x}}}+\ln x\frac{d}{dx}\left(e^{x^{e^{x}}}\right)\right\}$$

$$\Rightarrow\quad\frac{dy}{dx}=y\left\{\frac{e^{x^{e^{x}}}}{x}+\ln x\cdot e^{x^{e^{x}}}\cdot\frac{d}{dx}\left(x^{e^{x}}\right)\right\}=y\left\{\frac{e^{x^{e^{x}}}}{x}+\ln x\cdot e^{x^{e^{x}}}\cdot\frac{d}{dx}\left(e^{(e^{x}\cdot\ln x)}\right)\right\}$$

$$\Rightarrow\quad\frac{dy}{dx}=y\left\{\frac{e^{x^{e^{x}}}}{x}+\ln x\cdot e^{x^{e^{x}}}\cdot x^{e^{x}}\left\{\frac{e^{x}}{x}+\ln x\cdot e^{x}\cdot x^{x}(1+\ln x)\right\}\right\}$$

Q.9 If $y = (\sin x)^{\ln x}\,\text{cosec}\left(e^x(a+bx)\right)$ and $a+b = \frac{\pi}{2e}$ then the value of $\frac{dy}{dx}$ at $x = 1$ is

(A) $(\sin 1)\ln\sin(1)$ **(B)** 0 **(C)** $\ln\sin(1)$ **(D)** $1+\ln(\sin 1)$

Sol. **(C)**

$$y = (\sin x)^{\ln x}\,\text{cosec}\left(e^x(a+bx)\right)$$

$$\ln y = (\ln x)\ln(\sin x) - \ln\left(\sin\left(e^x(a+bx)\right)\right)$$

$$\frac{1}{y}\frac{dy}{dx} = \frac{1}{x}\ln(\sin x) + (\ln x)\cot x - \frac{\cos\left(e^x(a+bx)\right)\left(ae^x + bxe^x + be^x\right)}{\sin\left(e^x(a+bx)\right)}$$

$$\frac{1}{y}\frac{dy}{dx}\Big|_{\text{at } x=1} = \ln(\sin 1) + 0 - \frac{\cos\left(\frac{\pi}{2}\right)(a+2b)e}{\sin\left(\frac{\pi}{2}\right)} = \ln(\sin 1) + 0 - 0$$

$\left(\because a+b = \frac{\pi}{2e}\right)$ $\quad \therefore y = 1$ at $x = 1$

so $\left.\frac{dy}{dx}\right|_{\text{at } x=1} = \ln(\sin 1)\,y\big|_{\text{at } x=1} = \ln(\sin 1)\cdot 1 = \ln(\sin 1)$

Q.10 If $y = 2^{\log_2 x^{2x}} + \left(\tan\frac{\pi x}{4}\right)^{\frac{4}{\pi x}}$ then $\left.\frac{dy}{dx}\right|_{x=1}$ is

(A) 4 **(B)** $\frac{5}{2}$ **(C)** 3 **(D)** not defined

Sol. **(A)**

$$y = x^{2x} + \left(\tan\frac{\pi x}{4}\right)^{\frac{4}{\pi x}}$$

Let $u = x^{2x}$ so $\ln u = 2x\ln x$

$$\frac{1}{u}\frac{du}{dx} = 2(\ln x + 1) \Rightarrow \left.\frac{du}{dx}\right|_{x=1} = (1)(2) = 2$$

Let $v = \left(\tan\frac{\pi x}{4}\right)^{\frac{4}{\pi x}}$

$$\ln v = \frac{4}{\pi x}\ln\left(\tan\left(\frac{\pi x}{4}\right)\right)$$

$$\frac{1}{v}\frac{dv}{dx} = -\frac{4}{\pi x^2}\ln\left(\tan\left(\frac{\pi x}{4}\right)\right) + \frac{1}{x}\frac{\sec^2\left(\frac{\pi}{4}x\right)}{\tan\left(\frac{\pi}{4}x\right)}$$

$$\left.\frac{dv}{dx}\right|_{x=1} = 0 + 2$$

$$\Rightarrow \frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx} = 2 + 2 = 4$$

Q.11 **Let,** $y = \left(1+\frac{1}{x}\right)^x \cdot x^{\left(1+\frac{1}{x}\right)}$ **then y'(1) equals**

(A) ln(4e) **(B)** $\ln\left(\frac{4}{e}\right)$ **(C) 2 ln2 + 3** **(D) 2(ln 2 e)**

Sol. **(C)**

$$y = e^{\left(x\ln\left(1+\frac{1}{x}\right)\right)} \cdot e^{\left(\left(1+\frac{1}{x}\right)\ln x\right)} = e^{\left[x\ln\left(1+\frac{1}{x}\right)+\left(1+\frac{1}{x}\right)\ln x\right]}$$

$$y' = y\left[x\cdot\frac{1}{\left(1+\frac{1}{x}\right)}\cdot\left(\frac{-1}{x^2}\right)+\ln\left(1+\frac{1}{x}\right)+\left(1+\frac{1}{x}\right)\frac{1}{x}+\ln x\left(\frac{-1}{x^2}\right)\right]$$

$$y'(1) = e^{\ln 2}\cdot\left[\frac{-1}{2}+\ln 2+2\right] = 2\left[\frac{3}{2}+\ln 2\right]$$

$$y'(1) = 2\ln 2 + 3$$

Q.12 **Find** $\lim_{h\to 0}\frac{(e+h)^{\ln(e+h)} - e}{h}$.

Sol. $\lim_{h\to 0}\frac{(e+h)^{\ln(e+h)} - e^{\ln e}}{h} = \lim_{h\to 0}\frac{f(e+h)-f(e)}{h} = f'(e)$

$$\Rightarrow \quad f(x) = x^{\ln x} = e^{\ln x\cdot\ln x}$$

$$\Rightarrow \quad f'(x) = e^{(\ln x)^2}\cdot 2\ln x\cdot\frac{1}{x}$$

$$\Rightarrow \quad f'(e) = e\cdot 2\cdot\frac{1}{e} = 2$$

Q.13 $y = \frac{ax^2}{(x-a)(x-b)(x-c)} + \frac{bx}{(x-b)(x-c)} + \frac{c}{x-c} + 1;$

Prove that $\frac{1}{y}\cdot\frac{dy}{dx} = \frac{1}{x}\left(\frac{a}{a-x}+\frac{b}{b-x}+\frac{c}{c-x}\right)$.

Sol. $y = \dfrac{ax^2}{(x-a)(x-b)(x-c)} + \dfrac{bx}{(x-b)(x-c)} + \dfrac{c+x-c}{x-c}$

$\Rightarrow y = \dfrac{ax^2}{(x-a)(x-b)(x-c)} + \dfrac{bx}{(x-b)(x-c)} + \dfrac{(x-b)x}{(x-b)(x-c)}$

$\Rightarrow y = \dfrac{ax^2}{(x-a)(x-b)(x-c)} + \dfrac{x^2}{(x-b)(x-c)}$

$\Rightarrow y = \dfrac{ax^2}{(x-a)(x-b)(x-c)} + \dfrac{x^2(x-a)}{(x-a)(x-b)(x-c)}$

$\Rightarrow y = \dfrac{x^3}{(x-a)(x-b)(x-c)}$

Taking log on both sides, we get

$\Rightarrow \ln y = 3\ln x - \ln(x-a) - \ln(x-b) - \ln(x-c)$

Differentiating on both sides, we get

$\dfrac{1}{y}\cdot\dfrac{dy}{dx} = \dfrac{3}{x} - \dfrac{1}{x-a} - \dfrac{1}{x-b} - \dfrac{1}{x-c}$

$\Rightarrow \dfrac{1}{y}\cdot\dfrac{dy}{dx} = \left[\dfrac{1}{x} - \dfrac{1}{x-a}\right] + \left[\dfrac{1}{x} - \dfrac{1}{x-b}\right] + \left[\dfrac{1}{x} - \dfrac{1}{x-c}\right]$

$\Rightarrow \dfrac{1}{y}\cdot\dfrac{dy}{dx} = \dfrac{1}{x}\left[\dfrac{a}{a-x} + \dfrac{b}{b-x} + \dfrac{c}{c-x}\right]$

Hence proved.

Q.14 **Find** $f(x) = \lim\limits_{h\to 0} \dfrac{\left((\sin(x+h))^{\ln(x+h)} - (\sin x)^{\ln x}\right)}{h}$.

Sol. $f(x) = \lim\limits_{h\to 0} \dfrac{g(x+h) - g(x)}{h}$ (where $g(x) = (\sin x)^{\ln x}$)

$\Rightarrow f(x) = g'(x)$

$\Rightarrow f(x) = \dfrac{d}{dx}\left(e^{\ln x \cdot \ln\sin x}\right)$

$\Rightarrow f(x) = \left(e^{\ln x \cdot \ln\sin x}\right)\cdot\left(\dfrac{\ln\sin x}{x} + \ln x \cdot \dfrac{\cos x}{\sin x}\right)$

$\Rightarrow f(x) = (\sin x)^{\ln x}\left[\dfrac{\ln\sin x}{x} + \ln x \cdot \cot x\right]$

Q.15 $f = |x|^{|\sin x|}$ **k, find** $f'\left(-\frac{\pi}{4}\right)$.

Sol. For |x| and |sinx|, critical points are x = 0 and x = nπ respectively.

At $x = -\frac{\pi}{4}, f(x) = (-x)^{(-\sin x)}$

$\Rightarrow \quad f(x) = e^{-\sin x \cdot \ln(-x)}$

$\Rightarrow \quad f(x) = e^{-\sin x \cdot \ln(-x)}\left\{-\cos x \cdot \ln(-x) - \frac{1}{x} \cdot (-1) \cdot (-\sin x)\right\}$

$\Rightarrow \quad f'\left(-\frac{\pi}{4}\right) = \left(\frac{\pi}{4}\right)^{\left(\frac{1}{\sqrt{2}}\right)}\left\{-\frac{1}{\sqrt{2}}\ln\left(\frac{\pi}{4}\right) - \frac{1}{\left(\frac{-\pi}{4}\right)} \cdot \left(\frac{-1}{\sqrt{2}}\right)\right\} = \left(\frac{\pi}{4}\right)^{\frac{1}{\sqrt{2}}}\left\{\frac{-1}{\sqrt{2}}\ln\left(\frac{\pi}{4}\right) - \frac{4}{\sqrt{2}\pi}\right\}$

Note :

(i) If $\frac{dy}{dx}$ exists and $\frac{dy}{dx} \neq 0$, then $\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}$

or $\frac{dy}{dx} \cdot \frac{dx}{dy} = 1$ or $\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}}$. $\left[\frac{dx}{dy} \neq 0\right]$

Proof :

$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$ and $\frac{dx}{dy} = \lim_{\Delta y \to 0} \frac{\Delta x}{\Delta y}$

$\frac{dy}{dx} \cdot \frac{dx}{dy} = \left(\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}\right)\left(\lim_{\Delta y \to 0} \frac{\Delta x}{\Delta y}\right) = 1$

(ii) Product of $\frac{d^2y}{dx^2}$ and $\frac{d^2x}{dy^2}$ is not always 1.

Q.1 $\sqrt{x} + \sqrt{y} = 4$. **Find** $\frac{dx}{dy}$ **at y = 1.**

Sol.

Method-1

Differentiating w.r.t. y on both sides,

$$\frac{1}{2\sqrt{x}}\frac{dx}{dy} + \frac{1}{2\sqrt{y}} = 0 \qquad \ldots(1)$$

Now, at y = 1

$\sqrt{x} + \sqrt{1} = 4$

$\Rightarrow \sqrt{x} = 3$

$\Rightarrow x = 9$

Put x = 9 and y = 1 in equation (1), we get

$$\frac{1}{2 \cdot 3}\left.\frac{dx}{dy}\right|_{y=1} + \frac{1}{2 \cdot 1} = 0$$

$$\Rightarrow \frac{1}{6} \cdot \left.\frac{dx}{dy}\right|_{y=1} = \frac{-1}{2}$$

$$\Rightarrow \frac{dx}{dy}|_{y=1} = -3$$

Method-2

$$\sqrt{x} + \sqrt{y} = 4$$

Differentiating w.r.t. to x on both sides,

$$\frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}} \cdot \frac{dy}{dx} = 0$$

Put x = 9 and y = 1, we get

$$\frac{1}{2 \cdot 3} + \frac{1}{2 \cdot 1} \cdot \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{1}{2}\frac{dy}{dx} = \frac{-1}{6}$$

$$\Rightarrow \frac{dy}{dx} = \frac{-2}{6} = -\frac{1}{3}$$

$$\Rightarrow \frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}$$

$$\Rightarrow \frac{dx}{dy} = \frac{1}{\frac{-1}{3}} = -3$$

Q.2 $y = \sqrt{x}^{\sqrt{x}} \cdot e^{x^2}$, **find** $y'(1)$.

Sol. Let $f(x) = \sqrt{x}^{\sqrt{x}}$ and $g(x) = e^{x^2}$

$g'(x) = e^{x^2} \cdot 2x \Rightarrow g'(1) = 2e$

Now, $f(x) = \sqrt{x}^{\sqrt{x}}$

Taking logarithm on both sides,

$$\ln f(x) = \frac{1}{2}\sqrt{x}\ln x$$

Differentiating on both sides,

$$f'(x) = \sqrt{x}^{\sqrt{x}}\left[\frac{\ln x}{4\sqrt{x}} + \frac{1}{2\sqrt{x}}\right]$$

$$f'(1) = \frac{1}{2}$$

$\Rightarrow \quad y = f(x).g(x)$

Differentiating w.r.t. x on both sides,

$$y' = f'(x) \cdot g(x) + f(x) \cdot g'(x)$$

$\Rightarrow \quad y'(1) = f'(1) \cdot g(1) + f(1) \cdot g'(1)$

$\Rightarrow \quad y'(1) = \frac{1}{2} \cdot e + 1 \cdot 2e$

$\Rightarrow \quad y'(1) = \frac{5e}{2}$

Q.3 **If** $f(x) = y = \pi^2 + 2^x + x^2 + e^{\left(\frac{1}{x}\ln x\right)}$, **then find the slope of the perpendicular to the tangent on the graph of y = f(x) at x = 1.**

Sol. $y = \pi^2 + 2^x + x^2 + e^{\left(\frac{1}{x}\ln x\right)}$

$\Rightarrow \quad y' = 0 + 2^x \ln 2 + 2x + e^{\frac{\ln x}{x}}\left[\frac{-1}{x^2}\ln x + \frac{1}{x^2}\right]$

$\Rightarrow$ Slope of tangent at x = 1 on y = f(x) is,

$$m_1 = \frac{dy}{dx}\Big|_{x=1} = 2\ln 2 + 2 + 1[0 + 1] = 3 + 2\ln 2$$

$\Rightarrow$ Slope of perpendicular line to tangent at x = 1 is,

$$m_2 = \frac{-1}{m_1} = \frac{-1}{3 + 2\ln 2}$$

Parametric Differentiation

In some situation curves are represented by the equations e.g. $x = \sin t$ and $y = \cos t$

If $x = f(t)$ and $y = g(t)$ then

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{g'(t)}{f'(t)}$$

Note :

Some standard Substitutions

Expression	Substitution
$\sqrt{x^2 - a^2}$	$x = a\sin\theta$ or $a\cos\theta$
$\sqrt{a^2 + x^2}$	$x = a\tan\theta$ or $a\cot\theta$
$\sqrt{\frac{a+x}{a-x}}$ or $\sqrt{\frac{a-x}{a+x}}$	$x = a\cos\theta$ or $a\cos 2\theta$
$\sqrt{(a-x)(x-b)}$	$x = a\cos^2\theta + b\sin^2\theta$
$\sqrt{\frac{a-x}{x-b}}$	$x = a\cos^2\theta + b\sin^2\theta$

Q. **Find** $\frac{dy}{dx}$ **if**

(i) $x = a(\cos t + t\sin t)$ **and** $y = a(\sin t - t\cos t)$

(ii) $x = \frac{3at}{1+t^3}; y = \frac{3at^2}{1+t^3}$

(iii) $x = a\sec^2\theta; y = a\tan^2\theta$.

Sol. **(i)** $x = a(\cos t + t\sin t)$ and $y = a(\sin t - t\cos t)$

$$\frac{dx}{dt} = a(-\sin t + \sin t + t\cos t) = at\cos t$$

$$\frac{dy}{dt} = a(\cos t - \cos t + t\sin t) = at\sin t$$

$$\Rightarrow \frac{dy}{dx} = \tan t$$

(ii) $x = \frac{3at}{1+t^3}; y = \frac{3at^2}{1+t^3}$

$$\frac{dx}{dt} = \frac{3a\left((1+t^3).1 - t\left(3t^2\right)\right)}{\left(1+t^3\right)^2} = \frac{3a\left(1-2t^3\right)}{\left(1+t^3\right)^2}$$

$$\frac{dy}{dt} = \frac{3a\left((1+t^3)2t - t^2\left(3t^2\right)\right)}{\left(1+t^3\right)^2} = \frac{3a\left(2t-t^4\right)}{\left(1+t^3\right)^2}$$

$$\frac{dy}{dx} = \frac{\left(2-t^3\right)t}{\left(1-2t^3\right)}$$

(iii) $x = a\sec^2\theta; y = a\tan^2\theta$

$$\frac{dx}{d\theta} = 2a\sec\theta(\sec\theta\tan\theta)$$

$$\frac{dy}{d\theta} = 2a\tan\theta\sec^2\theta$$

$$\frac{dy}{dx} = 1$$

Q.6 **If** $x = \sec\theta - \cos\theta$ & $y = \sec^n\theta - \cos^n\theta$, **then show that** $\left(x^2+4\right)\left(\frac{dy}{dx}\right)^2 = n^2\left(y^2+4\right)$.

Sol. $x = \sec\theta - \cos\theta; y = \sec^n\theta - \cos^n\theta$

$$\frac{dx}{d\theta} = \sec\theta\tan\theta + \sin\theta = \sin\theta\left(\sec^2\theta+1\right)$$

$$\frac{dy}{d\theta} = n\sec^{n-1}\theta\sec\theta\tan\theta + n\cos^{n-1}\theta\sin\theta = n\sec^n\theta\tan\theta + n\cos^n\theta\tan\theta$$

$$\frac{dy}{d\theta} = n\tan\theta\left(\sec^n\theta + \cos^n\theta\right)$$

$$\frac{dy}{dx} = \frac{n\tan\theta\left(\sec^n\theta+\cos^n\theta\right)}{\sin\theta\left(\sec^2\theta+1\right)} = \frac{n\left(\sec^n\theta+\cos^n\theta\right)}{\cos\theta\left(\sec^2\theta+1\right)} = \frac{n\left(\sec^n\theta+\cos^n\theta\right)}{(\sec\theta+\cos\theta)}$$

$$\left(\frac{dy}{dx}\right)^2 = \frac{n^2\left(\sec^{2n}\theta+\cos^{2n}\theta+2\right)}{\left(\sec^2\theta+\cos^2\theta+2\right)} = \frac{n^2\left(y^2+4\right)}{\left(x^2+4\right)}$$

Q.3 **If** $x = a\sqrt{\cos 2t}\cos t$ **and** $y = a\sqrt{\cos 2t}\sin t$ **then, find** $\left.\frac{dy}{dx}\right|_{t=\frac{\pi}{6}}$.

Sol. $x = a\sqrt{\cos 2t}\cos t$ and $y = a\sqrt{\cos 2t}\sin t$

$$\frac{dx}{dt} = a\left(\sqrt{\cos 2t}(-\sin t) + \frac{\cos t(-2\sin 2t)}{2\sqrt{\cos 2t}}\right)$$

$$\frac{dx}{dt} = -a\left(\frac{\sin t\cos 2t + \cos t\sin 2t}{\sqrt{\cos 2t}}\right)$$

$$\frac{dx}{dt} = -a\frac{\sin 3t}{\sqrt{\cos 2t}} \Rightarrow \left.\frac{dx}{dt}\right|_{\frac{\pi}{6}} = -\sqrt{2}a$$

$$\frac{dy}{dt} = a\left(\sqrt{\cos 2t}\cos t + \frac{\sin t}{2\sqrt{\cos 2t}}(-2\sin 2t)\right) = a\left(\frac{\cos 3t}{\sqrt{\cos 2t}}\right) \Rightarrow \left.\frac{dy}{dt}\right|_{\frac{\pi}{6}} = 0$$

$$\left.\frac{dy}{dx}\right|_{\frac{\pi}{6}} = 0$$

Q.4 **For the curve represented parametrically indicate the relation between the parameter t and the angle α between the tangent to the given curve and the x-axis.**

(i) $\begin{cases} x = \cos t + t\sin t - \frac{t^2}{2}\cos t \\ y = \sin t - t\cos t - \frac{t^2}{2}\sin t \end{cases}$ **(ii)** $x = a\cos^3 t, y = a\sin^3 t$

Sol. **(i)** $x = \cos t + t\sin t - \frac{t^2}{2}\cos t$

$$\frac{dx}{dt} = -\sin t + t\cos t + \sin t + \frac{t^2}{2}\sin t - t\cos t = \frac{t^2\sin t}{2}$$

$$y = \sin t - t\cos t - \frac{t^2}{2}\sin t$$

$$\frac{dy}{dt} = \cos t - (\cos t - t\sin t) - \frac{1}{2}(2t\sin t + t^2\cos t) = \frac{-t^2\cos t}{2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{\left(\frac{dy}{dt}\right)}{\left(\frac{dx}{dt}\right)} = -\cot t = \tan\alpha \Rightarrow \tan\left(\frac{\pi}{2} - t\right) = \tan(-\alpha)$$

$$\Rightarrow \frac{\pi}{2} - t = -\alpha \Rightarrow t = \frac{\pi}{2} + \alpha$$

(ii) $x = a\cos^3 t \Rightarrow \frac{dx}{dt} = -3a\cos^2 t\sin t$

$y = a\sin^3 t \Rightarrow \frac{dy}{dt} = 3a\sin^2 t\cos t$

$$\Rightarrow \frac{dy}{dx} = \frac{\left(\frac{dy}{dt}\right)}{\left(\frac{dx}{dt}\right)} = \frac{3a\sin^2 t\cos t}{-3a\cos^2 t\sin t} = -\tan t = \tan\alpha$$

$$\Rightarrow \tan(\pi - t) = \tan\alpha \Rightarrow \pi - t = \alpha \Rightarrow t = \pi - \alpha$$

Q.5 **Find $\frac{dy}{dx}$ if $x = a(\theta - \sin\theta)$ and $y = a(1 - \cos\theta)$.**

Sol. We have,

$x = a(\theta - \sin\theta)$ and $y = a(1 - \cos\theta)$

$$\Rightarrow \frac{dx}{d\theta} = a(1 - \cos\theta) \text{ and } \frac{dy}{d\theta} = a\sin\theta$$

$$\Rightarrow \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{a\sin\theta}{a(1-\cos\theta)} = \frac{2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)}{2\sin^2\left(\frac{\theta}{2}\right)} = \cot\frac{\theta}{2}$$

Q.6 **If $x = a\sec^3\theta$ and $y = a\tan^3\theta$, find $\frac{dy}{dx}$ at $\theta = \frac{\pi}{3}$.**

Sol. We have $x = a\sec^3\theta$ and $y = a\tan^3\theta$

$$\frac{dx}{d\theta} = 3a\sec^3\theta\tan\theta \text{ and } \frac{dy}{d\theta} = 3a\tan^2\theta\sec^2\theta$$

$$\Rightarrow \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{3a\tan^2\theta\sec^2\theta}{3a\sec^3\theta\tan\theta} = \frac{\tan\theta}{\sec\theta} = \sin\theta$$

$$\Rightarrow \left.\frac{dy}{dx}\right|_{\theta=\frac{\pi}{3}} = \sin\frac{\pi}{3} = \frac{\sqrt{3}}{2}.$$

Q.7 **If $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x - y)$ then prove that $\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$.**

Sol. $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x - y)$

Put $x = \sin\alpha$, $y = \sin\beta$

$$\sqrt{1-\sin^2\alpha}+\sqrt{1-\sin^2\beta}=a(\sin\alpha-\sin\beta)$$

$$\cos\alpha+\cos\beta=a\left[2\cos\left(\frac{\alpha+\beta}{2}\right).\sin\left(\frac{\alpha-\beta}{2}\right)\right]$$

$$2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)=2a\cos\left(\frac{\alpha+\beta}{2}\right).\sin\left(\frac{\alpha-\beta}{2}\right)$$

$$\tan\left(\frac{\alpha-\beta}{2}\right)=\frac{1}{a}$$

$$\frac{\alpha-\beta}{2}=\tan^{-1}\frac{1}{a}$$

$$\sin^{-1}x-\sin^{-1}y=2\tan^{-1}\frac{1}{a}$$

$$\frac{1}{\sqrt{1-x^2}}-\frac{1}{\sqrt{1-y^2}}\frac{dy}{dx}=0\Rightarrow\frac{dy}{dx}=\sqrt{\frac{1-y^2}{1-x^2}}$$

DIFFERENTIATION OF ONE FUNCTION W.R.T. OTHER FUNCTION :

If $y = f(x)$ and $z = g(x)$ then derivative of $f(x)$ w.r.t. $g(x)$ is given by

$$\frac{dy}{dz}=\frac{dy}{dx}.\frac{dx}{dz}=\frac{f'(x)}{g'(x)}$$

$\Rightarrow$ Differential coefficient of $f(x)$ w.r.t. $g(x)$

$$=\frac{\text{derivative of } f(x) \text{ w.r.t. } x}{\text{derivative of } g(x) \text{ w.r.t.} x}=\frac{f'(x)}{g'(x)}$$

Q.1 **Find derivative of $(\ln x)^{\tan x}$ w.r.t. x^x.**

Sol. $u=(\ln x)^{\tan x}$; $v=x^x$

$$\ln(u)=\tan x\ln(\ln x)$$

$$\Rightarrow\frac{1}{u}\left(\frac{du}{dx}\right)=(\sec^2x)\ln(\ln x)+\tan x\left(\frac{1}{\ln x}.\frac{1}{x}\right)$$

$$\frac{du}{dx}=\frac{u\left((x\ln x)\ln(\ln x)\sec^2x+\tan x\right)}{(x\ln x)}$$

$$\ln v=x\ln x$$

$$\Rightarrow\frac{1}{v}\left(\frac{dv}{dx}\right)=(\ln x+1)$$

$$\frac{du}{dv}=\frac{(\ln x)^{\tan x}}{x^x}\left(\frac{x\ln x\ln(\ln x)\sec^2 x+\tan x}{(x\ln x)(\ln x+1)}\right)$$

Q.2 **Derivative of** $\cos^{-1}(2x^2-1)$ **w.r.t.** $\sqrt{1-x^2}$ **when** $x=\frac{1}{2}$.

Sol. $u=\cos^{-1}(2x^2-1);\ v=\sqrt{(1-x^2)}$

put $x=\cos\theta$ as $x=\frac{1}{2}$ so $\theta=\frac{\pi}{3}$

$$\frac{du}{dv}=\frac{\frac{du}{d\theta}}{\frac{dv}{d\theta}}=\frac{2}{\cos\theta}=4.$$

Q.3 **Define derivative of** $\sin^{-1}\left(\frac{2x}{1+x^2}\right)$ **w.r.t.** $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)\forall x\in R$.

Sol. Let $u=\sin^{-1}\left(\frac{2x}{1+x^2}\right)$ and $v=\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$

Let $x=\tan\theta$

$u=\sin^{-1}(\sin 2\theta)$ and $v=\cos^{-1}(\cos 2\theta)$

$$u=\begin{cases}2\theta, & -\frac{\pi}{2}\le 2\theta\le\frac{\pi}{2} \Rightarrow -\frac{\pi}{4}\le\theta\le\frac{\pi}{4} & \Rightarrow -1\le x\le 1\\ (-\pi-2\theta), & -\pi<2\theta\le\frac{-\pi}{2} \Rightarrow \frac{-\pi}{2}<\theta<\frac{-\pi}{4} & \Rightarrow -\infty<x<-1\\ (\pi-2\theta), & \frac{\pi}{2}\le 2\theta\le\pi \Rightarrow \frac{\pi}{4}<\theta<\frac{\pi}{2} & \Rightarrow 1<x<\infty\end{cases}$$

$v=\cos^{-1}(\cos 2\theta)$

$$v=\begin{cases}2\theta, & 0\le 2\theta\le\pi \Rightarrow 0\le\theta\le\frac{\pi}{2} & \Rightarrow 0\le x<\infty\\ -2\theta, & -\pi<2\theta<0 \Rightarrow \frac{-\pi}{2}<\theta<0 & \Rightarrow -\infty<x<0\end{cases}$$

$$\frac{du}{d\theta}=\begin{cases}2, & -1<x<1\\ -2, & 1<x<\infty\\ -2, & -\infty<x<-1\end{cases}$$

$$\frac{dv}{d\theta}=\begin{cases}2, 0<x<\infty\\ -2, -\infty<x<0\end{cases}$$

$$\frac{du}{dv}=\begin{cases}\left(\frac{-2}{-2}\right)=1, & -\infty<x<-1\\ \left(\frac{2}{-2}\right)=-1, & -1<x<0\\ \left(\frac{2}{2}\right)=1, & 0<x<1\\ \left(\frac{-2}{2}\right)=-1, & 1<x<\infty\end{cases}$$

$$\frac{du}{dv}=\begin{cases}1\ \forall\ x\in(-\infty,-1)\cup(0,1)\\ -1\ \forall\ x\in(-1,0)\cup(1,\infty)\end{cases}$$ not exists at x = –1, 0, 1

Q4. State True or False: Differential coefficient of $e^{\sin^{-1}x}$ w.r.t. $e^{-\cos^{-1}x}$ is independent of x.

Sol. Let $u=e^{\sin^{-1}x}$ and $v=e^{-\cos^{-1}x}$

$$u=e^{\frac{\pi}{2}-\cos^{-1}x} \qquad \left[\sin^{-1}x+\cos^{-1}x=\frac{\pi}{2}\right]$$

$$u=e^{\frac{\pi}{2}}e^{-\cos^{-1}x}=e^{\frac{\pi}{2}}v$$

So, $\frac{du}{dv}=e^{\frac{\pi}{2}}$.

Thus, **given statement is true.**

Q5. Find the derivative of f(tan x) w.r.t. g(sec x) at $x=\frac{\pi}{4}$, where $f'(1)=2 \text{ and } g'\left(\sqrt{2}\right)=4$.

Sol. Let $u=f(\tan x)$ and $v=g(\sec x)$

$$\Rightarrow \frac{du}{dx}=f'(\tan x)\sec^2 x \text{ and } \frac{dv}{dx}=g'(\sec x)\sec x\tan x$$

$$\Rightarrow \frac{du}{dv}=\frac{\frac{du}{dx}}{\frac{dv}{dx}}=\frac{f'(\tan x)\sec^2 x}{g'(\sec x)\sec x\tan x}$$

$$\Rightarrow \left[\frac{du}{dv}\right]_{x=\frac{\pi}{4}} = \frac{f'\left(\tan\frac{\pi}{4}\right)}{g'\left(\sec\frac{\pi}{4}\right)\sin\frac{\pi}{4}} = \frac{f'(1)\sqrt{2}}{g'\left(\sqrt{2}\right)} = \frac{2\sqrt{2}}{4} = \frac{1}{\sqrt{2}}$$

Q.6 **Differentiate ln(sinx) w.r.t. $\sqrt{\cos x}$.**

Sol. Let u = ln(sinx) and v= $\sqrt{\cos x}$

Then, $\frac{du}{dx} = \cot x$ and $\frac{dv}{dx} = -\frac{\sin x}{2\sqrt{\cos x}}$

$$\Rightarrow \frac{du}{dv} = \frac{\frac{du}{dx}}{\frac{dv}{dx}} = \frac{\cot x}{-\frac{\sin x}{2\sqrt{\cos x}}} = -2\sqrt{\cos x}\cot x \operatorname{cosec} x$$

Q.7 **Differentiate $\tan^{-1}\left(\frac{\sqrt{1-x^2}-1}{x}\right)$ w.r.t. $\tan^{-1}x$.**

Sol. Let $u = \tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$ and $v = \tan^{-1}x$.

Putting $x = \tan\theta$,

we get $u = \tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$

$$= \tan^{-1}\left(\frac{\sec\theta-1}{\tan\theta}\right) = \tan^{-1}\left(\frac{1-\cos\theta}{\sin\theta}\right) = \tan^{-1}\left(\tan\frac{\theta}{2}\right) = \frac{1}{2}\theta = \frac{1}{2}\tan^{-1}x$$

Thus, we have $u = \frac{1}{2}\tan^{-1}x$ and $v = \tan^{-1}x$.

$$\Rightarrow \frac{du}{dx} = \frac{1}{2}\times\frac{1}{1+x^2} \text{ and } \frac{dv}{dx} = \frac{1}{1+x^2}$$

$$\Rightarrow \frac{du}{dv} = \frac{\frac{du}{dx}}{\frac{dv}{dx}} = \frac{1}{2(1+x^2)}(1+x^2) = \frac{1}{2}$$

Implicit Functions

If the variable x and y connected by a relation of the form f(x, y) = 0 and it is not possible to express y as a function of x in the form $y = \phi(x)$, then such functions are said to be implicit functions. For example,

(i) $x^2 + xy + y^3 = 1$

(ii) $x + y + \sin(xy) = 1$

(iii) $x^y + y^x = 1$

(iv) $16^{x^2+y} + 16^{x+y^2} = 1$

DERIVATIVE OF IMPLICIT FUNCTION :

(i) In order to find $\frac{dy}{dx}$, in the case of implicit functions, we differentiate each term w.r.t. x regarding y as a function of x and then collect terms in $\frac{dy}{dx}$ together on one side to finally find $\frac{dy}{dx}$.

(ii) Corresponding to every curve represented by an implicit equation, there exist one or more explicit functions representing that equation. It can be shown that $\frac{dy}{dx}$ at any point on the curve remains the same whether the process of differentiation is done explicitly or implicitly.

$$\frac{d}{dx}f(y) = \frac{d}{dy}f(y)\cdot\frac{dy}{dx} = f'(y)\cdot\frac{dy}{dx}$$

A DIRECT FORMULA FOR IMPLICIT FUNCTIONS :

Let f(x, y) = 0. Take all the same terms of left side and put left side equal to f(x, y).

Then

$$\frac{dy}{dx} = -\frac{\text{diff. of f w.r.t. x keeping y as constant}}{\text{diff. of f w.r.t. y keeping x as constant}}$$

Q.1 **If $x^y = e^{x-y}$ then prove that $\frac{dy}{dx} = \frac{\ln x}{(1+\ln x)^2}$.**

Sol. $x^y = e^{x-y}$

$y\ln x = (x - y)$ ...(i)

Now differentiating with respect to x,

$$\frac{y}{x} + \ln x\cdot\frac{dy}{dx} = 1 - \frac{dy}{dx}$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{1-\frac{y}{x}}{1+\ln x} = \frac{x-y}{x(1+\ln x)} = \frac{y\ln x}{x(1+\ln x)} \qquad \left(\text{From (i)}, \ y = \frac{x}{1+\ln x}\right)$$

$$\therefore \quad \frac{dy}{dx} = \frac{\ln x}{(1+\ln x)^2}$$

Q.2 **If $\sin y = x \sin(a+y)$ then prove that $\frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin a}$.**

Sol. $\sin y = x \sin(a+y)$

Differentiating both sides w.r.t. x,

$$\cos y.\frac{dy}{dx} = \sin(a+y) + x\cos(a+y).\left(0+\frac{dy}{dx}\right).$$

$$\Rightarrow \quad \frac{dy}{dx}(\cos y - x\cos(a+y)) = \sin(a+y)$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{\sin(a+y)}{\cos y - x\cos(a+y)} = \frac{\sin(a+y)}{\cos y - \frac{\sin y}{\sin(a+y)}\cos(a+y)}$$

$$= \frac{\sin^2(a+y)}{\sin(a+y)\cos y - \sin y\cos(a+y)} = \frac{\sin^2(a+y)}{\sin a}$$

Q.3 **If $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \ldots + \infty}}}$ find $\frac{dy}{dx}(\sin x > 0)$.**

Sol. $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \ldots + \infty}}}$

$$\Rightarrow \quad y = \sqrt{\sin x + y}$$

$$\Rightarrow \quad y^2 - y = \sin x$$

$$\Rightarrow \quad 2y\frac{dy}{dx} - \frac{dy}{dx} = \cos x$$

$$\Rightarrow \quad \frac{dy}{dx} = \left(\frac{\cos x}{2y-1}\right)$$

Q.4 **Find derivative of $y = x^{x^{x^{\cdots}}}$.**

Sol. $y = x^{x^{x^{\cdots}}}$

$$\Rightarrow \quad y = x^y \ \Rightarrow \ln y = y\ln x \ \Rightarrow \frac{1}{y}.\frac{dy}{dx} = \frac{y}{x} + \ln x\frac{dy}{dx}$$

$$\Rightarrow \quad \frac{dy}{dx}\left(\frac{1}{y} - \ln x\right) = \frac{y}{x} \ \Rightarrow \frac{dy}{dx} = \frac{\frac{y}{x}}{\frac{(1-y\ln x)}{y}} = \frac{y^2}{x(1-y\ln x)}$$

Q.5 **Find derivative of** $y = (\ln x)^{(\ln x)^{(\ln x)^{\cdots}}}$.

Sol. $y = (\ln x)^y$

$\Rightarrow \ln y = y \ln(\ln x)$

$\Rightarrow \frac{1}{y}\cdot\frac{dy}{dx} = \frac{y}{x \ln x} + \ln\ln x \frac{dy}{dx}$

$\Rightarrow \frac{dy}{dx}\left(\frac{1}{y} - \ln(\ln x)\right) = \frac{y}{x \ln x}$

$\Rightarrow \frac{dy}{dx} = \frac{\frac{y}{x \ln x}}{\frac{\left(1 - y \ln(\ln x)\right)}{y}} = \frac{y^2}{x \ln x \left(1 - y \ln(\ln x)\right)}$.

Q.6 **If** $y = \cfrac{x}{1 + \cfrac{x}{2 + \cfrac{x}{1 + \cfrac{x}{2 + \ldots}}}}$ **then prove that** $y' = \frac{1}{1+y}$.

Sol. $y = \cfrac{x}{1 + \cfrac{x}{2 + y}}$

$\Rightarrow y = \frac{x(2+y)}{2 + y + x}$

$\Rightarrow 2y + y^2 + xy = 2x + yx$

$\Rightarrow 2y + y^2 = 2x$

$\Rightarrow 2\frac{dy}{dx} + 2y\frac{dy}{dx} = 2$ (By differentiating)

$\Rightarrow \frac{dy}{dx} = \frac{1}{1+y}$ **Hence Proved.**

Q.7 **If** $y = \cfrac{\sin x}{1 + \cfrac{\cos x}{1 + \cfrac{\sin x}{1 + \cos x \ldots}}}$ **then** $y'(0)$ **is**

(A) 0 **(B)** $\frac{1}{2}$ **(C) 1** **(D) non existent**

Sol. **(B)**

$y = \cfrac{\sin x}{1 + \cfrac{\cos x}{1 + \cfrac{\sin x}{1 + \cos x \ldots}}}$ at $x = 0, y = 0$

$$y = \frac{\sin x}{1+\frac{\cos x}{1+y}} \quad \Rightarrow \quad y = \frac{\sin x(1+y)}{1+y+\cos x}$$

$$\Rightarrow \; y + y^2 + y\cos x = \sin x + y\sin x$$

$$\Rightarrow \; \frac{dy}{dx} + 2y\frac{dy}{dx} + \cos x\frac{dy}{dx} - \sin x \cdot y = \cos x + y\cos x + \sin x \cdot \frac{dy}{dx}$$

$$\Rightarrow \; \frac{dy}{dx}(1+2y+\cos x - \sin x) = \cos x + y\cos x + y\sin x$$

$$\Rightarrow \; \frac{dy}{dx} = \frac{\cos x + y\cos x + y\sin x}{1+2y+\cos x - \sin x} \quad \Rightarrow \left.\frac{dy}{dx}\right|_{x=0} = \frac{1}{2}$$

Q.8 **A curve is described by the relation $\ln(x+y) = xe^y$. Find the tangent to the curve at (0, 1).**

Sol. $\ln(x+y) = xe^y$

Differentiating w.r.t. to x

$$\frac{1}{(x+y)}\left(1+\frac{dy}{dx}\right) = xe^y\frac{dy}{dx} + e^y$$

At point (0, 1),

$$\frac{1}{(0+1)}\left(1+\frac{dy}{dx}\right) = 0 + e^1$$

$$\Rightarrow \; \frac{dy}{dx}\Big|_{x=0,y=1} = e - 1$$

Equation of tangent is given as,

$y - 1 = (e - 1)(x - 0)$

$\Rightarrow$ **$(1 - e)x + y = 1$.**

Q.9 **If $y^5 + xy^2 + x^3 = 4x + 3$, then find $\frac{dy}{dx}$ at (2, 1).**

Sol. $y^5 + xy^2 + x^3 = 4x + 3$

Differentiating w.r.t. x

$$5y^4\frac{dy}{dx} + \left(x(2y)\frac{dy}{dx} + y^2\right) + 3x^2 = 4$$

$$\Rightarrow \; \frac{dy}{dx} = \frac{4 - 3x^2 - y^2}{(5y^4 + 2xy)}$$

$$\frac{dy}{dx}\Big|_{x=2,y=1} = \frac{4 - 3(2)^2 - (1)^2}{5(1)^4 + 2(2)(1)} = \frac{4 - 12 - 1}{5 + 4} = \frac{-9}{9} = -1.$$

Derivative of Inverse Function

Theorem :

Let the function f(x) and g(x) be inverse of each other.

Then $f(g(x)) = g(f(x)) = x$

$\therefore \quad f'(g(x))g'(x) = g'(f(x))f'(x) = 1$

$$\Rightarrow g'(f(x)) = \frac{1}{f'(x)}$$

Q.1 **If $y = f(x) = x^3 + x^5$ and g is the inverse of f, find $g'(2)$.**

Sol. $y = f(x) = x^3 + 5$

Let g(x) be the inverse of f(x) i.e., $g(x) = f^{-1}(x)$

$f(g(x)) = g(f(x)) = x$

Differentiating w.r.t. x,

$$f'(g(x)).g'(x) = 1$$

$$\Rightarrow \quad g'(2) = \frac{1}{f'(g(2))}$$

$$\Rightarrow \quad g(2) = f^{-1}(2) = y\,(\text{say})$$

$$\Rightarrow \quad f(y) = 2 \Rightarrow y^3 + y^5 = 2 \Rightarrow y = 1$$

$$\Rightarrow \quad g'(2) = \frac{1}{f'(1)} = \frac{1}{\left(3x^2 + 5x^4\right)_{x=1}} = \frac{1}{8}.$$

Q.2 **If $y = f(x) = x + x^3 + x^5 + x^7 + \ldots + x^{99}$ and g is the inverse of f, find $g'(50)$.**

Sol. $y = f(x) = x + x^3 + x^5 + \ldots + x^{99}$

$f(x) = 50$ at $x = 1$

$f'(x) = 1 + 3x^2 + 5x^4 + \ldots + 99x^{98}$

$f'(1) = 50^2 = 2500$

$g(x) = f^{-1}(x)$

$g(f(x)) = x$

We know that,

$$g'(f(x)) = \frac{1}{f'(x)}$$

$$g'(50) = \frac{1}{f'(1)} = \frac{1}{2500}$$

Q.3 **Let $f(x) = \exp(x^3 + x^2 + x)$ for any real number x, and let g be the inverse function for f. The value of $g'(e^3)$ is**

(A) $\frac{1}{6e^3}$ **(B)** $\frac{1}{6}$ **(C)** $\frac{1}{34e^{39}}$ **(D) 6**

Sol. $f(g(x)) = x$

$\Rightarrow f'(g(x))g'(x) = 1$

$\Rightarrow g'(x) = \frac{1}{f'(g(x))}$

$\Rightarrow g'(e^3) = \frac{1}{f'(g(e^3))}$

$\Rightarrow e^{x^3+x^2+x} = e^3 \quad \Rightarrow x = 1.$

$\Rightarrow g'(e^3) = \frac{1}{f'(1)} = \frac{1}{\left[e^{x^3+x^2+x}(3x^2+2x+1)\right]_{x=1}} = \frac{1}{e^3(3+2+1)} = \frac{1}{6e^3}.$

Q.4 **If g is the inverse of f and $f'(x) = \frac{1}{1+x^n}$, prove that $g'(x) = 1 + (g(x))^n$.**

Sol. $f(g(x)) = x$

$\Rightarrow f'(g(x))g'(x) = 1$

$\Rightarrow g'(x) = \frac{1}{f'(g(x))} = \frac{1}{\frac{1}{1+(g(x))^n}}$

$\Rightarrow g'(x) = 1 + (g(x))^n.$

Q.5 **If $f(x) = x^3 + e^{x/2}$ and $g(x) = f^{-1}(x)$. Find g' (1).**

Sol. $f(x) = x^3 + e^{x/2}$ and $f'(x) = 3x^2 + \frac{1}{2}e^{\frac{x}{2}}$

$g(x) = f^{-1}(x)$ $f'(0) = \frac{1}{2}$

$g(f(x)) = x$

$g'(f(x)) = \frac{1}{f'(x)}$

Now $f(x) = 1$ when $x = 0$

$g'(1) = \frac{1}{f'(0)}$

$g'(1) = \mathbf{2}$

Q.6 If $f(x) = -4e^{\frac{1-x}{2}} + \frac{x^3}{3} + \frac{x^2}{2} + x + 1$ and $g(x) = f^{-1}(x)$. Find $g'\left(-\frac{7}{6}\right)$.

Sol. $f(x) = -4e^{\frac{1-x}{2}} + \frac{x^3}{3} + \frac{x^2}{2} + x + 1$

$g(x) = f^{-1}(x)$ and $f'(x) = -4e^{\frac{1-x}{2}}\left(-\frac{1}{2}\right) + x^2 + x + 1$

$g(f(x)) = x$ and $f'(x) = 2e^{\frac{1-x}{2}} + x^2 + x + 1$

$g'(f(x)) = \frac{1}{f'(x)}$

$f(x) = -4e^{\frac{1-x}{2}} + \frac{x^3}{3} + \frac{x^2}{2} + x + 1 = \frac{7}{6} \Rightarrow x = 1$

$g'\left(-\frac{7}{6}\right) = \frac{1}{f'(1)} = \frac{1}{5}$

Q.7 If $f(x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \frac{x^5}{5}$ and $g(x) = f^{-1}(x)$. Find $g''(0)$.

Sol. $f(x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \frac{x^5}{5}$, f(x) = 0 when x = 0

$g(x) = f^{-1}(x) \Rightarrow g(f(x)) = x$

$g'(f(x)) = \frac{1}{f'(x)}$

$g''(f(x))f'(x) = -\frac{1}{f'^2(x)}f''(x) \Rightarrow g''(f(x)) = -\frac{f''(x)}{(f'(x))^3}$

$g'''(f(x))f'(x) = \frac{-(f'(x))^3 f'''(x) + f''(x)\,3(f'(x))^2 f''(x)}{(f'(x))^6}$

$g'''(0)\cdot 1 = \frac{-1\cdot 2 + 1\cdot 3\cdot 1}{1}$

$\Rightarrow$ $g'''(0) = 1$

DERIVATIVES OF INVERSE BY ELEMENTARY SUBSTITUTION :

Q.8 $y = \tan^{-1}\left(\frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}}\right)$

Sol. $y = \tan^{-1}\left(\frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}}\right)$

Let $x = \cos 2\theta \Rightarrow \theta = \frac{1}{2}\cos^{-1}(x)$

$$y = \tan^{-1}\left(\frac{\sqrt{1+\cos 2\theta} - \sqrt{1-\cos 2\theta}}{\sqrt{1+\cos 2\theta} + \sqrt{1-\cos 2\theta}}\right) = \tan^{-1}\left(\frac{\sqrt{2}\cos\theta - \sqrt{2}\sin\theta}{\sqrt{2}\cos\theta + \sqrt{2}\sin\theta}\right) = \tan^{-1}\left(\frac{1-\tan\theta}{1+\tan\theta}\right)$$

$$y = \tan^{-1}\left(\tan\left(\frac{\pi}{4} - \theta\right)\right) = \frac{\pi}{4} - \theta$$

$$y = \frac{\pi}{4} - \frac{1}{2}\cos^{-1}x$$

$$\therefore \frac{dy}{dx} = \frac{1}{2\sqrt{1-x^2}}$$

Q.9 $y = \tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$

Sol. $y = \tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$

Put $x = \tan\theta \therefore \theta = \tan^{-1}x$

$$y = \tan^{-1}\left(\frac{\sec\theta - 1}{\tan\theta}\right) = \tan^{-1}\left(\frac{1-\cos\theta}{\sin\theta}\right) = \tan^{-1}\left(\frac{2\sin^2\frac{\theta}{2}}{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}\right) = \tan^{-1}\left(\tan\frac{\theta}{2}\right)$$

$$\Rightarrow y = \frac{\theta}{2} = \frac{1}{2}\tan^{-1}x \Rightarrow \frac{dy}{dx} = \frac{1}{2(1+x^2)}$$

Q.10 **If $f(x) = \sin^{-1}\left(\frac{2^{x+1}}{1+2^{2x}}\right)$, find f'(0).**

Sol. $f(x) = \sin^{-1}\left(\frac{2^{x+1}}{1+2^{2x}}\right)$

Let $\theta = \tan^{-1}(2^x)$

$$\Rightarrow f(x) = \sin^{-1}\left(\frac{2\tan\theta}{1+\tan^2\theta}\right)$$

$$= \sin^{-1}(\sin 2\theta)$$

$$\Rightarrow f(x) = \begin{cases} 2\theta = 2\tan^{-1}(2^x), & x \in (-\infty, 0] \\ \pi - 2\theta = \pi - 2\tan^{-1}(2^x), & x \in (0, \infty] \end{cases}$$

$$f'(x) = \begin{cases} \left(\frac{2}{1+2^{2x}}\right)(2^x \ln 2), & x < 0 \\ \left(\frac{-2}{1+2^{2x}}\right)(2^x \ln 2), & x > 0 \end{cases}$$

At x = 0, LHD = ln2 and RHD = − ln2

∴ f'(0) does not exist.

N^{th} ORDER DERIVATIVES :

$y = f(x)$; the popular symbol used to denote the derivatives are

$$\frac{dy}{dx} = Dy = f'(x) = y_1 = y'.$$

Higher order derivatives are denotes as

$$\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d^2y}{dx^2} = D^2y = f''(x) = y_2 \text{ or } y'' \text{ etc.}$$

Q.1 **Find n^{th} order derivative of sinx, cosx, x^n, x^{n+1}.**

Sol.

$y = \sin x$	$y = \cos x$
$y_1 = \cos x$	$y_1 = -\sin x$
$y_2 = -\sin x$	$y_2 = -\cos x$
$y_3 = -\cos x$	$y_3 = \sin x$
$y_4 = \sin x$	$y_4 = \cos x$
If	If
$n = 4k$, $y_n = \sin x$	$n = 4k$, $y_n = \cos x$
$n = 4k + 1$, $y_n = \cos x$	$n = 4k + 1$, $y_n = -\sin x$
$n = 4k + 2$, $y_n = -\sin x$	$n = 4k + 2$, $y_n = -\cos x$
$n = 4k + 3$, $y_n = -\cos x$	**$n = 4k + 3$, $y_n = \sin x$**
$y = x^n$	$y = x^{n+1}$
$\frac{dy}{dx} = nx^{n-1}$	$\frac{dy}{dx} = (n + 1).x^n$
$\frac{d^2y}{dx^2} = n(n-1)x^{n-2}$	$\frac{d^2y}{dx^2} = (n + 1).\ n.\ x^{n-1}$
$\frac{d^3y}{dx^3} = n(n-1)(n-2)x^{n-3}$	:
:	:
:	:
$\mathbf{\frac{d^ny}{dx^n} = n(n-1)...1 = n!}$	$\mathbf{\frac{d^ny}{dx^n} = (n + 1)\cdot n.\ ...\ 2\cdot x}$

Note :

$\frac{d^2y}{dx^2}$ is double derivative of y w.r.t. x, i.e. $\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d^2y}{dx^2}$.

Q.2 **Find $\frac{d^2y}{dx^2}$ at $t = \frac{\pi}{4}$ if $y = \sin t$, $x = \cos t$.**

Sol. At $t = \frac{\pi}{4}$, $y = \frac{1}{\sqrt{2}}$, $x = \frac{1}{\sqrt{2}}$

$\frac{dx}{dt} = -\sin t$, $\frac{dy}{dt} = \cos t$

$\frac{dy}{dx} = -\cot t$ and $\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dt}\left(\frac{dy}{dx}\right)\frac{dt}{dx}$

So, $\frac{d^2y}{dx^2} = \operatorname{cosec}^2 t \cdot (-\operatorname{cosec} t) = -\operatorname{cosec}^3 t$

So, $\left.\frac{d^2y}{dx^2}\right|_{t=\frac{\pi}{4}} = -2\sqrt{2}$

Q.3 **Let $z = (\cos x)^5$ and $y = \sin x$. Then the value of $\frac{d^2z}{dy^2}$ at $x = \frac{2\pi}{9}$ is**

(A) $-\frac{1}{2}$ **(B)** $\frac{3}{2}$ **(C)** $\frac{5}{2}$ **(D)** $-\frac{3}{2}$

Sol. **(C)**

$z = (\cos x)^5$ and $y = \sin x$

$\frac{d^2z}{dy^2} = \frac{d}{dy}\left(\frac{dz}{dy}\right) = \frac{d}{dx}\left(\frac{dz}{dy}\right)\frac{dx}{dy}$

Now, $\frac{dz}{dx} = 5\cos^4 x(-\sin x)$ and $\frac{dy}{dx} = \cos x$

$\frac{dz}{dy} = 5\cos^3 x(-\sin x)$

$\frac{d}{dx}\left(\frac{dz}{dy}\right) = (-5)[\cos^3 x \cos x - \sin x \cdot 3\cos^2 x \sin x]$

So, $\frac{d^2z}{dy^2} = (-5)\cos^2 x\left[\cos^2 x - 3\sin^2 x\right]\frac{1}{\cos x}$

$\frac{d^2z}{dy^2} = (-5)\cos x\left[4\cos^2 x - 3\right]$

$\frac{d^2z}{dy^2} = (-5)[4\cos^3 x - 3\cos x] = (-5)\cos 3x$

$\left.\frac{d^2z}{dy^2}\right|_{x=\frac{2\pi}{9}} = (-5)\cos\frac{2\pi}{3} = \frac{5}{2}$

Q.4 **If $x = = a\cos\theta, y = b\sin\theta$, them find y''' in terms of θ.**

Sol. $x = = a\cos\theta, y = b\sin\theta$

$$\frac{dx}{d\theta} = = -a\sin\theta, \frac{dy}{d\theta} = b\cos\theta$$

$$\frac{dy}{dx} = -\frac{b}{a}\cot\theta$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{d\theta}\left(\frac{dy}{dx}\right)\frac{d\theta}{dx} = \frac{d}{d\theta}\left(-\frac{b}{a}\cot\theta\right)\cdot\frac{d\theta}{dx} = \frac{b}{a}\operatorname{cosec}^2\theta\cdot\frac{(-1)}{a\sin\theta}$$

$$\frac{d^2y}{dx^2} = \frac{-b}{a^2}\operatorname{cosec}^3\theta$$

$$\frac{d^3y}{dx^3} = \frac{d}{dx}\left(\frac{d^2y}{dx^2}\right) = \frac{d}{d\theta}\left(\frac{d^2y}{dx^2}\right)\frac{d\theta}{dx} = \frac{-b}{a^2}3\operatorname{cosec}^2\theta(-\operatorname{cosec}\theta\cdot\cot\theta)\frac{1}{-a\sin\theta}$$

$$\frac{d^3y}{dx^3} = \frac{3b}{a^3}\operatorname{cosec}^4\theta\cdot\cot\theta$$

Q.5 **If $2x = y^{\frac{1}{5}} + y^{-\frac{1}{5}}$ then $(x^2-1)y'' + xy' = \lambda y$. Find $\sqrt{\lambda}$.**

Sol. $2x = y^{\frac{1}{5}} + y^{-\frac{1}{5}}$

$$\left(y^{\frac{1}{5}}\right)^2 - 2xy^{\frac{1}{5}} + 1 = 0$$

$$y^{\frac{1}{5}} = \frac{2x \pm \sqrt{4x^2-4}}{2} = x + \sqrt{x^2-1}, x - \sqrt{x^2-1}$$

$$\frac{1}{5}\frac{1}{y^{\frac{4}{5}}}\frac{dy}{dx} = 1 + \frac{1(2x)}{2\sqrt{x^2-1}} = 1 + \frac{x}{\sqrt{x^2-1}} = \frac{x+\sqrt{x^2-1}}{\sqrt{x^2-1}}$$

$$y'\sqrt{x^2-1} = 5y$$

$$y''\sqrt{x^2-1} + y'\frac{x}{\sqrt{x^2-1}} = 5y'$$

$$y''(x^2-1) + y'x = 5(5y) = 25y$$

So, $\sqrt{\lambda} = 5$.

Q.6 **If $\sqrt{x^2+y^2} = e^{\sin^{-1}\left(\frac{y}{\sqrt{x^2+y^2}}\right)}$, then prove that $y'' = \frac{2(x^2+y^2)}{(x-y)^3}$.**

Sol. $\sqrt{x^2+y^2} = e^{\sin^{-1}\left(\frac{y}{\sqrt{x^2+y^2}}\right)}$

put $y = x\tan\theta$

we get $x\sec\theta = e^{\sin^{-1}\left(\frac{x\tan\theta}{x\sec\theta}\right)} = e^{\theta}$

$x = e^{\theta}\cos\theta$

$y = e^{\theta}\sin\theta$

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{e^{\theta}(\sin\theta + \cos\theta)}{e^{\theta}(\cos\theta - \sin\theta)} = \frac{\sin\theta + \cos\theta}{\cos\theta - \sin\theta}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{d\theta}\left(\frac{dy}{dx}\right)\frac{d\theta}{dx} = \frac{(\cos\theta - \sin\theta)(\cos\theta - \sin\theta) - (\sin\theta + \cos\theta)(-\sin\theta - \cos\theta)}{(\cos\theta - \sin\theta)^2} \Big/ e^{\theta}(\cos\theta - \sin\theta)$$

$$\frac{d^2y}{dx^2} = \frac{\left[(\cos\theta - \sin\theta)^2 + (\sin\theta + \cos\theta)^2\right]e^{2\theta}}{e^{\theta}(\cos\theta - \sin\theta)^3 \cdot e^{2\theta}} = \frac{2(x^2 + y^2)}{(x - y)^3}$$

Q.7 **For $f(x) = (x^2 + 2x + 2)^n$. Prove that**
(i) Coefficient of x is $n \cdot 2^n$. **(ii) Coefficient of x^2 is $n^2 2^{n-1}$.**

Sol. $f(x) = (x^2 + 2x + 2)^n$

$f'(x) = n(x^2 + 2x + 2)^{n-1}(2x + 2)$

$f'(0) = n \cdot 2^{n-1}(0 + 2) = \mathbf{n\,2^n}$ (coefficient of x)

$f''(x) = n(n - 1)(x^2 + 2x + 2)^{n-2}(2x + 2)^2 + n \cdot 2 \cdot (x^2 + 2x + 2)^{n-1}$

$f''(0) = n(n - 1) \cdot 2^{n-2} \cdot 2^2 + 2n \cdot 2^{n-1}$

$= \mathbf{n^2 2^n} = 2$ (coefficient of x^2)

$\Rightarrow$ Coefficient of $x^2 = n^2 2^{n-1}$

Q.8 **Starting with $\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}$. Prove that $\frac{d^2x}{dy^2} = -\frac{\frac{d^2y}{dx^2}}{\left(\frac{dy}{dx}\right)^3}$ and deduce that for the parabola $y^2 = 4ax$, $\frac{d^2y}{dx^2} \cdot \frac{d^2x}{dy^2} = -\frac{2a}{y^3}$.**

Sol. $$\frac{dx}{dy} = \frac{1}{\left(\frac{dy}{dx}\right)}$$

$$\frac{d^2x}{dy^2} = \frac{d}{dy}\left(\frac{1}{\left(\frac{dy}{dx}\right)}\right) = \frac{1}{dx}\left(\left(\frac{dy}{dx}\right)^{-1}\right)\cdot\frac{dx}{dy} = -1\left(\frac{dy}{dx}\right)^{-2}\left(\frac{d^2y}{dx^2}\right)\cdot\frac{1}{\left(\frac{dy}{dx}\right)} = \frac{-\left(\frac{d^2y}{dx^2}\right)}{\left(\frac{dy}{dx}\right)^3}$$

Given $y^2 = 4ax$

$$\Rightarrow \quad 2y\frac{dy}{dx} = 4a \Rightarrow y\frac{dy}{dx} = 2a \Rightarrow \frac{dy}{dx} = \frac{2a}{y} \Rightarrow \frac{d^2y}{dx^2} = \frac{-2a}{y^2}\left(\frac{dy}{dx}\right) = \frac{-4a^2}{y^3}$$

$$\left(\frac{d^2y}{dx^2}\right)\left(\frac{d^2x}{dy^2}\right) = \left(\frac{d^2y}{dx^2}\right)\left(\frac{-\left(\frac{d^2y}{dx^2}\right)}{\left(\frac{dy}{dx}\right)^3}\right) = \frac{-\left(\frac{-4a^2}{y^3}\right)^2}{\left(\frac{2a}{y}\right)^3} = \frac{-2a}{y^3}.$$

Q.9 **Use the substitution $x = \tan\theta$ to show that the equation, $\frac{d^2y}{dx^2} + \frac{2x}{1+x^2}\frac{dy}{dx} + \frac{y}{(1+x^2)^2} = 0$ changes to $\frac{d^2y}{d\theta^2} + y = 0$.**

Sol. $$\frac{d^2y}{dx^2} + \frac{2x}{1+x^2}\left(\frac{dy}{dx}\right) + \frac{y}{(1+x^2)^2} = 0 \qquad ...(1)$$

$$x = \tan\theta \Rightarrow dx = \sec^2\theta d\theta \Rightarrow \frac{d\theta}{dx} = \frac{1}{\sec^2\theta} = \cos^2\theta$$

$$\frac{dy}{dx} = \frac{dy}{d\theta}\left(\frac{d\theta}{dx}\right) = \frac{1}{\sec^2\theta}\left(\frac{dy}{d\theta}\right) = \cos^2\theta\left(\frac{dy}{d\theta}\right)$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{1}{\sec^2\theta}\cdot\frac{dy}{d\theta}\right) = \frac{d}{d\theta}\left(\cos^2\theta\frac{dy}{d\theta}\right)\cdot\frac{d\theta}{dx} = \left(\cos^2\theta\frac{d^2y}{d\theta^2} - 2\cos\theta\sin\theta\frac{dy}{d\theta}\right)\cos^2\theta$$

From equation (1), putting $x = \tan\theta$

$$\cos^2\theta\left(\cos^2\theta\frac{d^2y}{d\theta^2} - 2\cos\theta\sin\theta\frac{dy}{d\theta}\right) + \left(\frac{2\tan\theta}{1+\tan^2\theta}\right)\left(\cos^2\theta\frac{dy}{d\theta}\right) + \left(\frac{y}{(1+\tan^2\theta)^2}\right) = 0$$

$$\Rightarrow \quad \left(\frac{d^2y}{d\theta^2} - 2\tan\theta\frac{dy}{d\theta}\right) + \left(2\tan\theta\frac{dy}{d\theta}\right) + y = 0$$

$$\Rightarrow \quad \frac{d^2y}{d\theta^2} + y = 0$$

Q.10 **If $x \in (-\infty, -2) \cup (2, \infty)$, then the equation $y^3 - 3y + x = 0$ implicitly defines a unique real valued differentiable function $y = f(x)$. If $f\left(-10\sqrt{2}\right) = 2\sqrt{2}$ then $f''\left(-10\sqrt{2}\right)$ equals**

Sol. $y^3 - 3y + x = 0$

$3y^2y' - 3y' + 1 = 0$

$y' = \dfrac{1}{3 - 3y^2} \Rightarrow f'\left(-10\sqrt{2}\right) = -\dfrac{1}{21}$

$y'' = \dfrac{-1}{\left(3 - 3y^2\right)^2}\left(-6y\dfrac{dy}{dx}\right)$

$f''\left(-10\sqrt{2}\right) = \dfrac{6 \cdot 2\sqrt{2}\left(\dfrac{-1}{21}\right)}{7^2 \cdot 3^2} = \dfrac{-4\sqrt{2}}{7^3 3^2}$

Q.11 **$(x - a)^2 + (y - b)^2 = c^2$ $(c > 0)$ then** $\dfrac{\left[1 + \left(\dfrac{dy}{dx}\right)^2\right]^{3/2}}{\dfrac{d^2y}{dx^2}}$ can be **equal to**

(A) c **(B)** c^2 **(C)** c^3 **(D)** c^4

Sol. **(A)**

$(x - a)^2 + (y - b)^2 = c^2 \qquad (c > 0)$ ...(1)

Now differentiating w.r.t. to x

$2(x - a) + 2(y - b)y_1 = 0$

$\Rightarrow \quad (x - a) + (y - b)y_1 = 0$ ...(2)

$\Rightarrow \quad 1 + (y - b)y_2 + (y_1)^2 = 0$ ...(3)

$1 + (y_1)^2 = 1 + \dfrac{(x - a)^2}{(y - b)^2}$ (from (2))

$1 + (y_1)^2 = \dfrac{c^2}{(y - b)^2}$

$\dfrac{\left(1 + (y_1)^2\right)^{\frac{3}{2}}}{y_2} = \dfrac{\left(\dfrac{c^2}{(y - b)^2}\right)^{\frac{3}{2}}}{\dfrac{-\left(1 + (y_1)^2\right)}{(y-b)}}$ (from (3))

$= \dfrac{\left|\dfrac{c^3}{(y - b)^3}\right|}{-\dfrac{c^2}{(y - b)^3}} = \pm c$.

Q.12 **If $y = \left(\frac{1}{x}\right)^x$ then prove that $y_2(1) = 0$ i.e. $\left.\frac{d^2y}{dx^2}\right|_{x=1} = 0$.**

Sol. $y = x^{-x} \Rightarrow \ln y = -x\ln x$

$$\frac{1}{y}\left(\frac{dy}{dx}\right) = -(1+\ln x) \Rightarrow \frac{dy}{dx} = -y(1+\ln x) \text{ so } \left.\frac{dy}{dx}\right|_{x=1} = -y$$

$$\frac{d^2y}{dx^2} = -\left(\frac{dy}{dx}\right)(1+\ln x) - \frac{y}{x}$$

$$\left.\frac{d^2y}{dx^2}\right|_{x=1} = -\left.\frac{dy}{dx}\right|_{x=1} - y = y - y = 0$$

Q.13 **If $y = e^{\tan^{-1}x}$ then prove that $(1+x^2)y''' + (4x-1)y'' + 2y' = 0$**

Sol. $y = e^{\tan^{-1}x}$

$$\frac{dy}{dx} = \frac{e^{\tan^{-1}x}}{(1+x^2)}$$

$(1 + x^2)y' = e^{\tan^{-1}x} = y$

$(1 + x^2)y'' + 2xy' = y'$

$(1 + x^2)y''' + 2xy'' + 2xy'' + 2y' = y''$

$(1 + x^2)y''' + (4x - 1)y'' + 2y' = 0$

Q.14 **If $y = \sin(m \sin^{-1}x)$, then show that $(1-x^2)y_{n+2} = (2n+1)xy_{n+1} + (n^2 - m^2)y_n$**

Sol. $y = \sin(m\sin^{-1}x)$ then $(1-x^2)y_{n+2} = (2n+1)xy_{n+1} + (n^2-m^2)y_n$

Now, differentiating w.r.t. to x

$$\frac{dy}{dx} = \cos(m\sin^{-1}x)\frac{m}{\sqrt{(1-x^2)}}$$

$$y_1 = \pm\sqrt{(1-y^2)}\frac{m}{\sqrt{(1-x^2)}}$$

Now, squaring we get

$$y_1^2 = m^2\frac{1-y^2}{1-x^2}$$

$$\Rightarrow (1-x^2)y_1^2 = m^2(1-y^2)$$

$$\Rightarrow (1-x^2)2y_1y_2 - 2xy_1^2 = -m^2 2yy_1$$

$\Rightarrow (1-x^2)y_2 - xy_1 + m^2y = 0$

So for $n = 0; (1-x^2)y_2 - xy_1 + m^2y = 0$

Let the equation $(1-x^2)y_{n+2} = (2n+1)xy_{n+1} + (n^2 - m^2)y_n$ is true for any n then again differentiating.

$$(1-x^2)y_{n+3} - 2xy_{n+2} = (2n+1)xy_{n+2} + (2n+1)y_{n+1} + (n^2 - m^2)y_{n+1}$$

$$\Rightarrow (1-x^2)y_{n+3} - 2xy_{n+2} - (2n+1)xy_{n+2} = \left[(n+1)^2 - m^2\right]y_{n+1}$$

$$\Rightarrow (1-x^2)y_{n+3} = \left[2(n+1)+1\right]xy_{n+2} + \left[(n+1)^2 - m^2\right]y_{n+1}$$

Note :

A homogeneous equation of degree n represents 'n' straight lines passing through the origin, hence,

$$\frac{dy}{dx} = \frac{y}{x} \text{ and } \frac{d^2y}{dx^2} = 0$$

e.g. If $x^3 + 3x^2y - 6xy^2 + 2y^3 = 0$,

then $\left.\frac{d^2y}{dx^2}\right|_{(1,1)} = 0$

DEDUCTION OF NEW IDENTITIES BY DIFFERENTIATING A GIVEN IDENTITY :

Q.1 **If** $\cos\frac{x}{2}\cdot\cos\frac{x}{2^2}\cdot\cos\frac{x}{2^3}\cdot\ldots\cdot\cos\frac{x}{2^n} = \frac{\sin x}{2^n \sin\left(\frac{x}{2^n}\right)}$, **then prove that**

$$\sum_{r=1}^{n}\frac{1}{2^r}\tan\frac{x}{2^r} = \frac{1}{2^n}\cot\frac{x}{2^n} - \cot x$$

Sol. $$\cos\frac{x}{2}\cdot\cos\frac{x}{2^2}\cdot\cos\frac{x}{2^3}\cdot\ldots\cdot\cos\frac{x}{2^n} = \frac{\sin x}{2^n \sin\left(\frac{x}{2^n}\right)} \quad \ldots(1)$$

Taking logarithm of both sides,

$$\log\left(\cos\frac{x}{2}\right) + \log\left(\cos\frac{x}{2^2}\right) + \log\left(\cos\frac{x}{2^3}\right) + \ldots + \log\left(\cos\frac{x}{2^n}\right) = \log(\sin x) - \log\left(2^n \sin\left(\frac{x}{2^n}\right)\right)$$

Differentiating w.r.t. to x,

$$-\left(\frac{1}{2}\frac{\sin\frac{x}{2}}{\cos\frac{x}{2}}+\frac{1}{2^2}\frac{\sin\left(\frac{x}{2^2}\right)}{\cos\left(\frac{x}{2^2}\right)}+\ldots+\frac{1}{2^n}\frac{\sin\left(\frac{x}{2^n}\right)}{\cos\left(\frac{x}{2^n}\right)}\right)=\cot x-\frac{1}{2^n\sin\left(\frac{x}{2^n}\right)}\cdot\cos\left(\frac{x}{2^n}\right)$$

So, $\sum_{r=1}^{n}\frac{1}{2^r}\tan\left(\frac{x}{2^r}\right)=\frac{1}{2^n}\cot\left(\frac{x}{2^n}\right)-\cot x$ **Hence proved.**

Q.2 **Prove that :** $C_0+2C_1+3C_2+\ldots+(n+1)C_n=(n+2)2^{n-1}$

Sol. We know that, $(1+x)^n=C_0+C_1x+C_2x^2+\ldots+C_nx^n$ ...(1)

Now multiply x on both sides of equation (1)

$x(1+x)^n=C_0x+C_1x^2+C_2x^3+\ldots+C_nx^{n+1}$

Differentiating equation (1) w.r.t. to x,

$(1+x)^n+nx(1+x)^{n-1}=C_0+2C_1x+3C_2x^2+\ldots+(n+1)C_nx^n$

Put x = 1

$2^n+n2^{n-1}=C_0+2C_1+3C_2+\ldots+(n+1)C_n$

$2^{n-1}(n+2)=C_0+2C_1+3C_2+\ldots+(n+1)C_n$ **Hence proved.**

Q.3 **Prove that :** $(n-1)^2\cdot C_1+(n-3)^2\cdot C_3+(n-5)^2\cdot C_5+\ldots=n(n+1)2^{n-3}$

Sol. We know that,

$(1+x)^n=C_0+C_1x+C_2x^2+\ldots+C_nx^n$

$(1+x)^n=C_0x^n+C_1x^{n-1}+C_2x^{n-2}+\ldots+C_n$

Differentiating on both sides w.r.t. x,

$n(1+x)^{n-1}=n\cdot C_0x^{n-1}+(n-1)C_1x^{n-2}+(n-2)C_2x^{n-3}+\ldots$ ...(1)

Now multiply x on both sides of equation (1),

$nx(1+x)^{n-1}=n\cdot C_0x^n+(n-1)C_1x^{n-1}+(n-2)C_2x^{n-2}+\ldots$

Differentiating on both sides w.r.t. x,

$n(1+x)^{n-1}+nx(n-1)(1+x)^{n-2}=n^2C_0x^{n-1}+(n-1)^2C_1x^{n-2}+\ldots$ ...(2)

Assuming n as an even number,

Put x = – 1 in equation (2),

$0=-n^2\cdot C_0+(n-1)^2\cdot C_1-(n-2)^2\cdot C_2+\ldots$ ...(3)

Put x = 1 in equation (2),

$n(2)^{n-1}+n(n-1)(2)^{n-2}=n^2\cdot C_0+(n-1)^2\cdot C_1+(n-2)^2\cdot C_2+\ldots$ ...(4)

Adding (3) and (4),

$$n(2)^{r-1} + n(n-1)(2)^{n-2} = 2\left((n-1)^2 \cdot C_1 + (n-3)^2 \cdot C_3 + (n-5)^2 \cdot C_5 + \ldots\right)$$

$$2n(2)^{n-3} + (n^2 - n)(2)^{n-3} = (n-1)^2 \cdot C_1 + (n-3)^2 \cdot C_3 + (n-5)^2 \cdot C_5 + \ldots$$

$$(n-1)^2 \cdot C_1 + (n-3)^2 \cdot C_3 + (n-5)^2 \cdot C_5 + \ldots = n(n+1)2^{r-3}$$

Similarly, assuming n as an odd number,
Put x = – 1 in equation (2),

$$0 = n^2 C_0 - (n-1)^2 \cdot C_1 + (n-2)^2 \cdot C_2 - \ldots \quad \ldots(5)$$

Put x = 1 in equation (2),

$$n(2)^{r-1} + n(n-1)(2)^{n-2} = n^2 \cdot C_0 + (n-1)^2 \cdot C_1 + (n-2)^2 \cdot C_2 + \ldots \quad \ldots(6)$$

(6) – (5) gives :

$$n(2)^{r-1} + n(n-1)(2)^{n-2} = 2\left(n^2 \cdot C_0 + (n-2)^2 \cdot C_2 + (n-4)^2 \cdot C_4 + \ldots\right)$$

$$(n-1)^2 \cdot C_1 + (n-3)^2 \cdot C_3 + (n-5)^2 \cdot C_5 + \ldots = n(n+1)2^{n-3}$$ **Hence proved.**

Q.4 **If 0 < x < 1, prove that** $\frac{1}{1+x} + \frac{2x}{1+x^2} + \frac{4x^3}{1+x^4} + \ldots.\infty = \frac{1}{1-x}$

Sol. Let $y = (1+x)(1+x^2)(1+x^4)\ldots$ ... (1)

multiply and divide by (1 – x)

$$y = \frac{(1-x)(1+x)(1+x^2)(1+x^4)\ldots}{(1-x)}$$

$$y = \frac{(1-x^2)(1+x^2)(1+x^4)\ldots}{(1-x)}$$

$$y = \frac{1}{1-x} \quad \ldots (2)$$

$$(1+x)(1+x^2)(1+x^4) \ldots\ldots = \frac{1}{1-x}$$

$\ln(1+x) + \ln(1+x^2) + \ln(1+x^4)\ldots = -\ln(1-x)$

by differentiating, we get

$$\frac{1}{1+x} + \frac{2x}{1+x^2} + \frac{4x^3}{1+x^4} + \ldots = \frac{1}{1-x}$$ **Hence proved.**

Derivative of Functions Expressed In The Determinant Form

Let $F(x)=\begin{vmatrix} f & g & h \\ u & v & w \\ l & m & n \end{vmatrix}$ where all functions are differentiable then

$$D'(x)=\begin{vmatrix} f' & g' & h' \\ u & v & w \\ l & m & n \end{vmatrix}+\begin{vmatrix} f & g & h \\ u' & v' & w' \\ l & m & n \end{vmatrix}+\begin{vmatrix} f & g & h \\ u & v & w \\ l' & m' & n' \end{vmatrix}$$

This result may be proved functions by first principle and the same operation can also be done column wise.

Q.1 **If** $f(x)=\begin{vmatrix} \cos(x+x^2) & \sin(x+x^2) & -\cos(x+x^2) \\ \sin(x-x^2) & \cos(x-x^2) & \sin(x-x^2) \\ \sin 2x & 0 & \sin 2x^2 \end{vmatrix}$ **then find** $f'(x)$.

Sol. $f(x)=\begin{vmatrix} \cos(x+x^2) & \sin(x+x^2) & -\cos(x+x^2) \\ \sin(x-x^2) & \cos(x-x^2) & \sin(x-x^2) \\ \sin 2x & 0 & \sin 2x^2 \end{vmatrix}$

Expending the determinant

$$f(x)=\cos(x+x^2)\cos(x-x^2)\sin 2x^2-\sin(x+x^2)\left(\sin(x-x^2)\right)\left(\sin 2x^2-\sin 2x\right)$$
$$+\cos(x+x^2)\cos(x-x^2)\sin 2x$$
$$=\sin 2x^2\cos 2x+\sin 2x\cos(2x^2)$$
$$=\sin(2x^2+2x)$$
$$f'(x)=\cos(2x^2+2x)\,2(2x+1)$$

Q.2 **If α be a repeated root of a quadratic equation f(x) = 0 and A(x), B(x), C(x) be the polynomials of degree 3, 4 and 5 respectively, then show that** $\begin{vmatrix} A(x) & B(x) & C(x) \\ A(\alpha) & B(\alpha) & C(\alpha) \\ A'(\alpha) & B'(\alpha) & C'(\alpha) \end{vmatrix}$ **is divisible by f(x), where dash denotes the derivative.**

Sol. $g(x)=\begin{vmatrix} A(x) & B(x) & C(x) \\ A(\alpha) & B(\alpha) & C(\alpha) \\ A'(\alpha) & B'(\alpha) & C'(\alpha) \end{vmatrix}$ and $g'(x)=\begin{vmatrix} A'(x) & B'(x) & C'(x) \\ A(\alpha) & B(\alpha) & C(\alpha) \\ A'(\alpha) & B'(\alpha) & C'(\alpha) \end{vmatrix}$

so $g(\alpha)=0$ and $g'(\alpha)=0$

so α is a repeated root of g(x).

so $g(x)=(x-\alpha)^2 h(x)=kf(x)\,h(x)$, where k is a number

so g(x) is divisible by f(x).

Q.3 **If f, g and h are differentiable functions of x and** $D=\begin{vmatrix} f & g & h \\ (xf)' & (xg)' & (xh)' \\ (x^2f)'' & (x^2g)'' & (x^2h)'' \end{vmatrix}$ **prove that** $D'=\begin{vmatrix} f & g & h \\ f' & g' & h' \\ (x^3f'')' & (x^3g'')' & (x^3h'')' \end{vmatrix}$.

Sol. $D=\begin{vmatrix} f & g & h \\ (xf)' & (xg)' & (xh)' \\ (x^2f)'' & (x^2g)'' & (x^2h)'' \end{vmatrix}$

$D=\begin{vmatrix} f & g & h \\ f+xf' & g+xg' & h+xh' \\ 2f+4xf'+x^2f'' & 2g+4xg'+x^2g'' & 2h+4xh'+x^2h'' \end{vmatrix}$

After doing row and column operation, this determinant simplifies to

$D=\begin{vmatrix} f & g & h \\ xf' & xg' & xh' \\ x^2f'' & x^2g'' & x^2h'' \end{vmatrix}=x^3\begin{vmatrix} f & g & h \\ f' & g' & h' \\ f'' & g'' & h'' \end{vmatrix}$

$D=\begin{vmatrix} f & g & h \\ f' & g' & h' \\ x^3f'' & x^3g'' & x^3h'' \end{vmatrix}$

$D'=\begin{vmatrix} f & g & h \\ f' & g' & h' \\ (x^3f'')' & (x^3g'')' & (x^3h'')' \end{vmatrix}$

Hence proved.

Note :

If (x – r) is a factor of the polynomial $f(x) = a_n x^r + a_{n-1}x^{n-1} + a_{r-2}x^{r-2} + \ldots + a_0$ repeated m times where $1 \le m \le n$ then r is a root of the equation $f'(x) = 0$ repeated (m – 1) times.

Q.4 **If** $f(x) = \begin{vmatrix} \cos x & \sin x & \cos x \\ \cos 2x & \sin 2x & 2\cos 2x \\ \cos 3x & \sin 3x & 3\cos 3x \end{vmatrix}$ **then** $f'\left(\frac{\pi}{2}\right)$ **equals**

Sol.

$$f'(x) = \begin{vmatrix} -\sin x & \cos x & -\sin x \\ \cos 2x & \sin 2x & 2\cos 2x \\ \cos 3x & \sin 3x & 3\cos 3x \end{vmatrix} + \begin{vmatrix} \cos x & \sin x & \cos x \\ -2\sin 2x & 2\cos 2x & -4\sin 2x \\ \cos 3x & \sin 3x & 3\cos 3x \end{vmatrix} + \begin{vmatrix} \cos x & \sin x & \cos x \\ \cos 2x & \sin 2x & 2\cos 2x \\ -3\sin 3x & 3\cos 3x & -9\sin 3x \end{vmatrix}$$

$$f'\left(\frac{\pi}{2}\right) = \begin{vmatrix} -1 & 0 & -1 \\ -1 & 0 & -2 \\ 0 & -1 & 0 \end{vmatrix} + \begin{vmatrix} 0 & 1 & 0 \\ 0 & -2 & 0 \\ 0 & -1 & 0 \end{vmatrix} + \begin{vmatrix} 0 & 1 & 0 \\ -1 & 0 & -2 \\ 3 & 0 & 9 \end{vmatrix}$$

$$f'\left(\frac{\pi}{2}\right) = 4$$

Q.5 **Given that,** $f(x) = \begin{vmatrix} x^3 & \sin x & \cos x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$ **where p is a constant. If f'' (0) = 0, find p.**

Sol.

$$f(x) = \begin{vmatrix} x^3 & \sin x & \cos x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$$

$$f'(x) = \begin{vmatrix} 3x^2 & \cos x & -\sin x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$$

$$f''(x) = \begin{vmatrix} 6x & -\sin x & -\cos x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$$

$$f''(0) = \begin{vmatrix} 0 & 0 & -1 \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix} = 0$$

$\Rightarrow 6p^2 + p = 0 \Rightarrow p = 0$ or $p = -\frac{1}{6}$

Q.6 **If f, g and h are polynomial of degree 2, then prove that $\phi(x) = \begin{vmatrix} f & g & h \\ f' & g' & h' \\ f'' & g'' & h'' \end{vmatrix}$ is constant polynomial.**

Sol. $\phi(x) = \begin{vmatrix} f & g & h \\ f' & g' & h' \\ f'' & g'' & h'' \end{vmatrix}$

$$\phi'(x) = \begin{vmatrix} f' & g' & h' \\ f' & g' & h' \\ f'' & g'' & h'' \end{vmatrix} + \begin{vmatrix} f & g & h \\ f'' & g'' & h'' \\ f'' & g'' & h'' \end{vmatrix} + \begin{vmatrix} f & g & h \\ f' & g' & h' \\ f''' & g''' & h''' \end{vmatrix}$$

$\because \quad f''', g''', h''' = 0$

So $\phi'(x) = 0$

Hence $\phi(x)$ is a constant polynomial.

Q.7 **If $y = \begin{vmatrix} \sin x & \cos x & \sin x \\ \cos x & -\sin x & \cos x \\ x & 1 & 1 \end{vmatrix}$ then $\frac{dy}{dx}$ =?**

Sol. $y = \begin{vmatrix} \sin x & \cos x & \sin x \\ \cos x & -\sin x & \cos x \\ x & 1 & 1 \end{vmatrix}$

$$y' = \begin{vmatrix} \cos x & -\sin x & \cos x \\ \cos x & -\sin x & \cos x \\ x & 1 & 1 \end{vmatrix} + \begin{vmatrix} \sin x & \cos x & \sin x \\ -\sin x & -\cos x & -\sin x \\ x & 1 & 1 \end{vmatrix} + \begin{vmatrix} \sin x & \cos x & \sin x \\ \cos x & -\sin x & \cos x \\ 1 & 0 & 0 \end{vmatrix}$$

$y' = \cos^2 x + \sin^2 x = 1$

Q.8 **If $f(x) = \begin{vmatrix} x^r & n! & 2 \\ \cos x & \cos\frac{n\pi}{2} & 4 \\ \sin x & \sin\frac{n\pi}{2} & 8 \end{vmatrix}$. Find $\frac{d^r}{dx^n}(f(x))_{x=0}$.**

Sol. $f(x) = \begin{vmatrix} x^n & n! & 2 \\ \cos x & \cos\frac{n\pi}{2} & 4 \\ \sin x & \sin\frac{n\pi}{2} & 8 \end{vmatrix}$

$f'(x) = \begin{vmatrix} nx^{n-1} & n! & 2 \\ -\sin x & \cos\frac{n\pi}{2} & 4 \\ \cos x & \sin\frac{n\pi}{2} & 8 \end{vmatrix}$

:

:

$f^r(x)|_{x=0} = \begin{vmatrix} n! & n! & 2 \\ \cos\frac{n\pi}{2} & \cos\frac{n\pi}{2} & 4 \\ \sin\frac{n\pi}{2} & \sin\frac{n\pi}{2} & 8 \end{vmatrix} = 0$

Q.9 **If** $f(x) = \begin{vmatrix} x+a^2 & ab & ac \\ ab & x+b^2 & bc \\ ac & bc & x+c^2 \end{vmatrix}$ **then prove that** $f'(x) = 3x^2 + 2x(a^2 + b^2 + c^2)$.

Sol. $f(x) = \begin{vmatrix} x+a^2 & ab & ac \\ ab & x+b^2 & bc \\ ac & bc & x+c^2 \end{vmatrix}$

$f'(x) = \begin{vmatrix} 1 & 0 & 0 \\ ab & x+b^2 & bc \\ ac & bc & x+c^2 \end{vmatrix} + \begin{vmatrix} x+a^2 & ab & ac \\ 0 & 1 & 0 \\ ac & bc & x+c^2 \end{vmatrix} + \begin{vmatrix} x+a^2 & ab & ac \\ ab & x+b^2 & bc \\ 0 & 0 & 1 \end{vmatrix}$

$f'(x) = (x + b^2)(x + c^2) - b^2c^2 + (x + a^2)(x + c^2) - a^2c^2 + (x + a^2)(x + b^2) - a^2b^2$

$f'(x) = 3x^2 + 2x(a^2 + b^2 + c^2)$

Note :

The derivative of an even differentiable function is an odd function, and the derivative of an odd differentiable function is an even function.

Proof :

Let f(x) be even function.	**Let f(x) be odd function.**
$f(-x) = f(x)$	$f(-x) = -f(x)$
$f'(-x)(-1) = f'(x)$	$f'(-x)(-1) = -f'(x)$
$\Rightarrow f'(-x) = -f'(x)$	$\Rightarrow f'(-x) = f'(x)$
$\Rightarrow f'(x)$ is an odd function.	$\Rightarrow f'(x)$ is an even function.

L' HOSPITAL'S RULE $\left(\frac{0}{0} \text{ or } \frac{\infty}{\infty}\right)$

If f(x) and g(x) are two functions such that

(i) $\lim\limits_{x\to a} f(x) = 0$ and $\lim\limits_{x\to a} g(x) = 0$

(ii) f and g are derivable/ continuous at $x = a$

i.e. $\lim\limits_{x\to a} f(x) = f(a) = 0; \lim\limits_{x\to a} g(x) = g(a) = 0$

(iii) $f'(x)$ and $g'(x)$ are continuous at $x = a$, then

$$\lim_{x\to a}\frac{f(x)}{g(x)} = \lim_{x\to a}\frac{f'(x)}{g'(x)} = \lim_{x\to a}\frac{f''(x)}{g''(x)}$$ till the

indeterminate form vanishes.

Q.1 **Find a and b if** $\lim\limits_{x\to 0}\frac{x(1+a\cos x) - b\sin x}{x^3} = 1$

Sol. Let $L = \lim\limits_{x\to 0}\frac{x(1+a\cos x) - b\sin x}{x^3}$ $\left(\frac{0}{0}\right)$form

$$L = \lim_{x\to 0}\frac{(1+a\cos x) - ax\sin x - b\cos x}{3x^2} \quad \left(\frac{0}{0}\right)\text{form}$$

As $D' = 0 \Rightarrow N' = 1 + a - 0 - b = 0$

$\Rightarrow b - a = 1$ ...(1)

$$L = \lim_{x\to 0}\frac{-a\sin x - a(x\cos x + \sin x) + b\sin x}{6x}$$

$$= \lim_{x\to 0}\frac{1}{6}\left[-a\left(\frac{\sin x}{x}\right) - a\cos x - \frac{a\sin x}{x} + \frac{b\sin x}{x}\right] = \frac{1}{6}(-a - a - a + b) = 1$$

$\Rightarrow -3a + b = 6$ ...(2)

From equation (1) and (2),

$$a = \frac{-5}{2}, b = \frac{-3}{2}$$

Q.2 **Evaluate** $\lim\limits_{x\to 0}\dfrac{1+\sin x-\cos x+\ln(1-x)}{x\tan^2 x}$

(A) $-\dfrac{1}{2}$ **(B)** $-\dfrac{1}{3}$ **(C)** $\dfrac{1}{6}$ **(D) Do Not Exist**

Sol. **(A)**

$$\lim_{x\to 0}\frac{1+\sin x-\cos x+\ln(1-x)}{x\tan^2 x}$$

$$=\lim_{x\to 0}\frac{1+\sin x-\cos x+\ln(1-x)}{x^3\left(\frac{\tan^2 x}{x^2}\right)}$$

$$=\lim_{x\to 0}\frac{1+\sin x-\cos x+\ln(1-x)}{x^3}\qquad \left(\frac{0}{0}\right)\text{ form}$$

$$=\lim_{x\to 0}\frac{\cos x+\sin x-\frac{1}{1-x}}{3x^2}\qquad \left(\frac{0}{0}\right)\text{ form}$$

$$=\lim_{x\to 0}\frac{-\sin x+\cos x-\frac{1}{(1-x)^2}}{6x}\qquad \left(\frac{0}{0}\right)\text{ form}$$

$$=\lim_{x\to 0}\frac{-\cos x-\sin x-\frac{2}{(1-x)^3}}{6}=\frac{-1-0-2}{6}=-\frac{1}{2}$$

Q.3 **Evaluate** $\lim\limits_{x\to 0}\dfrac{\log_{\sec\left(\frac{x}{2}\right)}(\cos x)}{\log_{\sec x}\left(\cos\left(\frac{x}{2}\right)\right)}$

(A) 1 **(B) 16** **(C) 4** **(D) 2**

Sol. **(B)**

$$\lim_{x\to 0}\frac{\log_{\sec\left(\frac{x}{2}\right)}(\cos x)}{\log_{\sec x}\cos\left(\frac{x}{2}\right)}=\lim_{x\to 0}\frac{\left(\frac{\ln\cos x}{\ln\sec\frac{x}{2}}\right)}{\left(\frac{\ln\cos\frac{x}{2}}{\ln\sec x}\right)}=\lim_{x\to 0}\left(\frac{\ln\cos x}{\ln\cos\frac{x}{2}}\right)^2$$

$$= \left(\lim_{x\to 0} \frac{\ln \cos x}{\ln \cos\left(\frac{x}{2}\right)} \right)^2 \left(\frac{0}{0}\right) \text{ form}$$

$$= \left(\lim_{x\to 0} \frac{-\tan x}{\frac{-1}{2}\tan\frac{x}{2}} \right)^2 = \left(\lim_{x\to 0} 4\frac{\frac{\tan x}{x}}{\frac{\tan\frac{x}{2}}{\left(\frac{x}{2}\right)}} \right)^2 = 4^2 = 16$$

Q.4 **Evaluate** $\lim_{x\to 1} \frac{x^x - x}{x - 1 - \ln x}$.

Sol. $\lim_{x\to 1} \frac{x^x - x}{x - 1 - \ln x} = \lim_{x\to 1} \frac{e^{x\ln x} - x}{x - 1 - \ln x} \left(\frac{0}{0}\right)$ from

$$\lim_{x\to 1} \frac{e^{x\ln x}(1+\ln x) - 1}{\left(1 - \frac{1}{x}\right)} \left(\frac{0}{0}\right) \text{ from}$$

$$= \lim_{x\to 1} \frac{e^{x\ln x}(1+\ln x)^2 + e^{x\ln x}\cdot\frac{1}{x} - 0}{\left(\frac{1}{x^2}\right)} \left(\frac{0}{0}\right) \text{ form } = 1 + 1 - 0 = 2$$

Q.5 **Evaluate** $\lim_{x\to 0} \frac{x\cos x - \ln(1+x)}{x^2}$.

Sol. $\lim_{x\to 0} \frac{x\cos x - \ln(1+x)}{x^2} \left(\frac{0}{0}\right)$ form

$$= \lim_{x\to 0} \frac{(\cos x - x\sin x) - \frac{1}{1+x}}{2x} \left(\frac{0}{0}\right) \text{ from}$$

$$= \lim_{x\to 0} \frac{-\sin x - (\sin x + x\cos x) + \frac{1}{(1+x)^2}}{2} = \frac{-0 - (0+0) + 1}{2} = \frac{1}{2}$$

Q.6 $\lim_{x\to 0} \frac{e^x \sin x - x - x^2}{x^2 + x - \ln(1-x)}$

Sol. $L = \lim_{x\to 0} \frac{e^x \sin x - x - x^2}{x^2 + x - \ln(1-x)} \left(\frac{0}{0}\right)$ from

$$= \lim_{x\to 0} \frac{e^x(\cos x + \sin x) - 1 - 2x}{2x + 1 + \frac{1}{(1-x)}}$$

$$= \frac{1-1-0}{0+1+1} = 0.$$

Q.7 **Evaluate** $\lim_{x\to 0} \frac{1 - \cos x \cdot \cos 2x \cdot \cos 3x}{x^2}$

Sol. $\lim_{x\to 0} \frac{1 - \cos x \cos 2x \cos 3x}{x^2} = \lim_{x\to 0} \frac{1 - e^{(\ln\cos x + \ln\cos 2x + \ln\cos 3x)}}{x^2}$

$$= \lim_{x\to 0} \frac{-e^{\ln\cos x + \ln\cos 2x + \ln\cos 3x} \cdot (-\tan x - 2\tan 2x - 3\tan 3x)}{2x} = \frac{e^0}{2}(1+4+9) = 7.$$

Note :

If $\lim_{x\to a} f(x) \to \infty$ and $\lim_{x\to a} g(x) \to \infty$

then also $\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$

Q.8 **Evaluate** $\lim_{x\to 0^+} (\operatorname{cosec} x)^{\frac{1}{\ln x}}$

Sol. $L = \lim_{x\to 0^+} (\operatorname{cosec} x)^{\frac{1}{\ln x}}$

$$\ln L = \lim_{x\to 0^+} \frac{\ln \operatorname{cosec} x}{\ln x} \left(\frac{\infty}{\infty}\right) \text{form}$$

$$= \lim_{x\to 0^+} \frac{\frac{1}{\operatorname{cosec} x}(-\operatorname{cosec} x \cot x)}{\left(\frac{1}{x}\right)} = \lim_{x\to 0^+} -\left(\frac{x}{\tan x}\right) = -1.$$

$$\Rightarrow L = e^{-1}$$

Q.10 **Evaluate** $\lim_{x\to 0^+} x^m (\log x)^n, m, n \in N.$

Sol. $\lim_{x\to 0^+} x^m (\log x)^n = \lim_{x\to 0^+} \frac{(\log x)^n}{x^{-m}} \left(\frac{\infty}{\infty} \text{form}\right)$

$$= \lim_{x\to 0^+} \frac{n(\log x)^{(n-1)} \frac{1}{x}}{-mx^{-m-1}}$$ (using L' hospital's Rule)

$$= \lim_{x\to 0^+} \frac{n(\log x)^{(n-1)}}{-mx^{-m}} \left(\frac{\infty}{\infty} \text{form}\right)$$

$$= \lim_{x\to 0^+} \frac{n(n-1)(\log x)^{(n-2)}\frac{1}{x}}{(-m^2)x^{-m-1}}$$ (using L'hospital's Rule)

$$= \lim_{x\to 0^+} \frac{n(n-1)(\log x)^{(n-2)}}{m^2 x^{-m}} \left(\frac{\infty}{\infty}\text{form}\right)$$

$$= \lim_{x\to 0^+} \frac{n!}{(-m)^n x^{-m}} = 0$$ (differentiating N' and D' n times)

Q.10 **Evaluate** $\lim_{x\to 0}\left(\frac{1}{x}\right)^{\tan x}$

Sol. $L = \lim_{x\to 0}\left(\frac{1}{x}\right)^{\tan x}$

$$\ln L = \lim_{x\to 0} \tan x(-\ln x) = \lim_{x\to 0} \frac{-\ln x}{\cot x} \left(\frac{\infty}{\infty}\right)\text{form}$$

$$\lim_{x\to 0} \frac{-\frac{1}{x}}{(-\text{cosec}^2 x)} = \lim_{x\to 0}\left(\frac{\sin x}{x}\right)\cdot \sin x = 0$$

$L = 1$

Q.11 **Evaluate** $\lim_{x\to 0} x^x$

Sol. $L = \lim_{x\to 0} x^x$

$$\ln L = \lim_{x\to 0} x\ln x = \lim_{x\to 0} \frac{\ln x}{\left(\frac{1}{x}\right)} \left(\frac{\infty}{\infty}\right) \text{form}$$

$$= \lim_{x\to 0} \frac{\frac{1}{x}}{\left(\frac{-1}{x^2}\right)} = \lim_{x\to 0}(-x) = 0$$

$L = 1$

Q.12 **Evaluate** $\lim_{x\to 0^+} (\sin x)^{\frac{1}{x}}$.

Sol. $y = \lim_{x\to 0^+} (\sin x)^{\frac{1}{x}}$

It is of the form $0^\infty = 0$

$\Rightarrow y = 0$

Q.13 **Let f(x) be a differentiable function and f''(0) = 2, then find**

$$\lim_{x\to 0}\frac{2f(x)-3f(2x)+f(4x)}{x^2}$$

Sol.

$$\lim_{x\to 0}\frac{2f(x)-3f(2x)+f(4x)}{x^2}$$

$$=\lim_{x\to 0}\frac{2f'(x)-3\cdot 2f'(2x)+4f'(4x)}{2x}$$

$$=\lim_{x\to 0}\frac{2f''(x)-12f''(2x)+16f''(4x)}{2}$$

$$=\frac{2\cdot 2-12\cdot 2+16\cdot 2}{2}$$

$$=\frac{36-24}{2}=\frac{12}{2}=6$$